Dar Alfikr Schools – Mathematics Department

AP Calculus AB

Solutions Booklet

Worked solutions to every "Check yourself" and unit practice item

Free-response solutions in scoring-guide style

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Unit 1 Solutions: Limits and Continuity

Topic check-yourself answers

Topics 1.1–1.16

Topic 1.1 check. (1) $s(3) = 3(9) + 3 = 30$, $s(1) = 3 + 1 = 4$; average velocity $= \dfrac{30-4}{3-1} = 13$ m/s.
(2) $\dfrac{s(2+h)-s(2)}{h} = \dfrac{(2+h)^2+(2+h)-6}{h} = \dfrac{5h+h^2}{h} = 5+h$. For $h = 0.1$: 5.1; for $h = 0.01$: 5.01. The averages approach 5 (the instantaneous velocity at $t = 2$).

Topic 1.2 check. (1) Left: $\lim\limits_{x\to 2^-}(x^2-1) = 3$. Right: $\lim\limits_{x\to 2^+}(7-x) = 5$. The one-sided limits differ, so $\lim\limits_{x\to 2} f(x)$ does not exist ($f(2) = 6$ is irrelevant).
(2) $\lim\limits_{x\to 0^-}\dfrac{1}{x} = -\infty$ (negative inputs give large negative outputs); $\lim\limits_{x\to 0}\dfrac{1}{x^2} = \infty$.
(3) 7: the one-sided limits agree and $g(4)$ plays no role.

Topic 1.3 check. (1) $\lim\limits_{x\to 1^+} h(x) = 2$ and $\lim\limits_{x\to -2} h(x) = 2$; sum $= 4$.
(2) Near $x = 0$, $h(x) = x + 4 \to 4$; as $x \to \infty$ the right-hand branch $\dfrac{8}{(x-3)^2} \to 0$. So 4 and 0.
(3) Both branches rise without bound at $x = 3$, so $\lim\limits_{x\to 3} h(x) = \infty$: it is infinite (the limit is not a real number, but it is $+\infty$, not $-\infty$).

Topic 1.4 check. (1) Both sides approach 10: $\lim\limits_{x\to 5} f(x) \approx 10$.
(2) As $x \to 1^-$, $g \to 3$; as $x \to 1^+$, $g \to -3$. The one-sided limits differ, so $\lim\limits_{x\to 1} g(x)$ does not exist.
(3) At $x = \frac{1}{n}$, $\sin\frac{\pi}{x} = \sin(n\pi) = 0$ for every n, so the table shows only 0. But at $x = \frac{2}{4n+1}$ the function equals $\sin\frac{(4n+1)\pi}{2} = 1$. Both sequences approach 0, so the limit does not exist; the table sampled the wrong points.

Topic 1.5 check. (1) $2(4) + 3(-2) = 2$; $\dfrac{4}{-2} = -2$; $(-2)^2 = 4$.
(2) The denominator at $x = -1$ is $5 \neq 0$: $\dfrac{(-1)^3 + 2(-1)}{1+4} = \dfrac{-3}{5} = -\dfrac{3}{5}$.
(3) $\dfrac{\sqrt{16-7}}{4-2} = \dfrac{3}{2}$.

Topic 1.6 check. (1) $\dfrac{(x-5)(x+5)}{(x-5)(x+2)} = \dfrac{x+5}{x+2} \to \dfrac{10}{7}$.
(2) Multiply by the conjugate: $\dfrac{(x+25)-25}{x(\sqrt{x+25}+5)} = \dfrac{1}{\sqrt{x+25}+5} \to \dfrac{1}{10}$.
(3) $\dfrac{\tan 4x}{8x} = \dfrac{4}{8}\cdot\dfrac{\sin 4x}{4x}\cdot\dfrac{1}{\cos 4x} \to \dfrac{1}{2}\cdot 1\cdot 1 = \dfrac{1}{2}$.

Topic 1.7 check. (1) Substitution gives $\frac{0}{0}$. Multiply by $\dfrac{\sqrt{x+7}+3}{\sqrt{x+7}+3}$: $\dfrac{(x-2)(\sqrt{x+7}+3)}{x+7-9} = \sqrt{x+7}+3 \to 6$.
(2) At $x = -3$ the numerator is -2; for $x > -3$ the denominator is a small positive number, so the quotient is a large negative number: the limit is $-\infty$.
(3) Numerator must vanish at $x = 2$: $4 + 2k - 10 = 0$, $k = 3$. Then $\dfrac{x^2+3x-10}{x-2} = \dfrac{(x+5)(x-2)}{x-2} = x + 5 \to 7$.

Topic 1.8 check. (1) For $x \neq 0$, $-1 \le \cos\dfrac{1}{x^2} \le 1$, so $-|x| \le x\cos\dfrac{1}{x^2} \le |x|$. Both bounds $\to 0$, so by the Squeeze Theorem the limit is 0.
(2) Both bounds tend to 3 as $x \to 3$ ($3 - 0$ and $3 + 0$), so the limit is 3.
(3) $\dfrac{\sin 7x}{\sin 3x} = \dfrac{\sin 7x}{7x}\cdot\dfrac{3x}{\sin 3x}\cdot\dfrac{7}{3} \to 1\cdot 1\cdot\dfrac{7}{3} = \dfrac{7}{3}$.

Topic 1.9 check. (1) $\dfrac{x^2-9}{x-3} = x+3$ for $x \neq 3$, so the limit is 6, matching the table (values $5.9, 5.99, 5.999$ and $6.001, 6.01, 6.1$).
(2) Yes. The graph approaches height 2 from both sides, with an open circle at $(a, 2)$, and has a separate filled dot at $(a, 7)$.

Topic 1.10 check. (1) $f(x) = \dfrac{1}{x+2}$ for $x \neq 2$. At $x = 2$: limit $\dfrac{1}{4}$ exists, $f(2)$ undefined: *removable*. At $x = -2$: unbounded: *infinite* discontinuity.
(2) No. Left limit 4, right limit 6: a *jump* of size $6 - 4 = 2$.
(3) Infinite discontinuity: $\lim\limits_{x\to 0} \dfrac{1}{x^2} = \infty$, regardless of $f(0) = 5$.

Topic 1.11 check. (1) $\lim\limits_{x\to 1} f(x) = \lim\limits_{x\to 1}(x+1) = 2$, but $f(1) = 3 \neq 2$, so f is not continuous at 1.
(2) $g(3) = 5$, $\lim\limits_{x\to 3^-} g = 6 + k$, so $6 + k = 5$, $k = -1$.
(3) $f(2)$ is undefined (condition 1 fails; the limit also fails to exist).

Topic 1.12 check. (1) f: need $x + 5 \geq 0$, $x \neq 2$: continuous on $[-5, 2) \cup (2, \infty)$. g: need $x - 3 > 0$: continuous on $(3, \infty)$.
(2) On $[1, 3]$ the function is not continuous (undefined at 2), so the theorem does not apply (and there is no maximum or minimum). On $[3, 6]$ it is continuous on a closed interval, so an absolute maximum ($f(3) = 1$) and minimum ($f(6) = \frac{1}{4}$) exist.
(3) $f(2) = 6 - 2 = 4$; $\lim\limits_{x\to 2^-} x^2 = 4$; $\lim\limits_{x\to 2^+} (6 - x) = 4$. All equal, so f is continuous at 2.

Topic 1.13 check. (1) $\dfrac{x^2-16}{x-4} = x + 4 \to 8$, so define $f(4) = 8$.
(2) $\dfrac{x^3-1}{x-1} = x^2 + x + 1 \to 3$, so $k = 3$.
(3) $f(2) = 9$; left limit $2a + 1 = 9 \Rightarrow a = 4$; right limit $4 + b = 9 \Rightarrow b = 5$.

Topic 1.14 check. (1) Numerator $\to 5 > 0$. For $x \to 3^+$ the denominator is a small positive number: $+\infty$. For $x \to 3^-$ it is a small negative number: $-\infty$.
(2) $f(x) = \dfrac{x-1}{(x-3)(x+2)}$. At $x = 3$ the numerator is $2 \neq 0$ and at $x = -2$ it is $-3 \neq 0$, so the vertical asymptotes are $x = 3$ and $x = -2$.
(3) $\lim\limits_{x\to 0^+} \ln x = -\infty$; $\lim\limits_{x\to(\pi/2)^+} \tan x = -\infty$.

Topic 1.15 check. (1) Divide by x^2: $\dfrac{4 - 1/x}{1/x^2 - 2} \to \dfrac{4}{-2} = -2$.
(2) As $x \to \infty$: $\dfrac{5x+1}{|x|\sqrt{1 + 1/x^2}} \to 5$. As $x \to -\infty$: $|x| = -x$ gives $\to -5$. Asymptotes $y = 5$ and $y = -5$.
(3) First limit: numerator degree $3 > 2$, so $\lim = \infty$ (no horizontal asymptote). Second: degree $1 < 2$, so $\lim = 0$.

Topic 1.16 check. (1) $f(x) = x^5 + x - 1$ is a polynomial, hence continuous on $[0, 1]$; $f(0) = -1 < 0 < 1 = f(1)$. By the IVT there is c in $(0, 1)$ with $f(c) = 0$.
(2) $f(2) = 5 > 0 > -3 = f(6)$ and f is continuous, so yes, $f(c) = 0$ for some c in $(2, 6)$. No: 7 is not between -3 and 5, so the IVT does not guarantee $f(c) = 7$.
(3) Sign changes: $f(1) = 2 > 0 > -1 = f(3)$ gives a zero in $(1, 3)$; $f(3) < 0 < f(4)$ gives one in $(3, 4)$. On $[4, 8]$ both values are 4 (no sign change). Minimum: 2.

Part A solutions

Multiple choice, no calculator

1. Answer: B.
1. The curve approaches 3 as $x \to 0^-$ (the line $y = x + 3$) and as $x \to 0^+$ (the arch $3 + \frac{x(3-x)}{3}$). The open circle at $(0, 3)$ marks where both pieces are heading; the dot $F(0) = 1$ is irrelevant.
Why not the others: (A) reads $F(0) = 1$, confusing value with limit. (C) 4 is the value at $x = 3$. (D) takes the

displaced dot to mean the limit does not exist.

2. Answer: C.
1. As $x \to 3^-$, $F(x) \to 3$ (arch returns to $y = 3$). 2. As $x \to 3^+$, $F(x) = 1 + \dfrac{3}{x-2} \to 4$.
3. $3 \neq 4$, so the limit does not exist.
Why not the others: (A) uses only the left limit. (B) uses only the right limit (which also equals $F(3)$). (D) averages the one-sided limits.

3. Answer: A.
1. At $x = 3$: $\lim\limits_{x\to 3^+} F = 4 = F(3)$, so F is right-continuous, but $\lim\limits_{x\to 3^-} F = 3 \neq 4$, so it is not continuous.
2. At $x = 0$: $\lim_{x\to 0^+} F = 3 \neq F(0) = 1$ (not right-continuous). At $x = -2$: $F(-2) = 1 = \lim F$ (continuous). At $x = 5$: $F(5) = 1 + \frac{3}{3} = 2 = \lim F$ (continuous).
Why not the others: (B) is a removable discontinuity that fails both one-sided tests. (C) and (D) are points of continuity.

4. Answer: D.
1. Substitution gives $\frac{0}{0}$. 2. $\dfrac{x^2+x-2}{x+2} = \dfrac{(x+2)(x-1)}{x+2} = x - 1$ for $x \neq -2$. 3. $\lim\limits_{x\to -2}(x-1) = -3$.
Why not the others: (A) treats $\frac{0}{0}$ as 0. (B) a sign slip, $-2-1$ read as 3. (C) treats $\frac{0}{0}$ as "does not exist".

5. Answer: A.
$\dfrac{\sin 4x}{2x} = 2 \cdot \dfrac{\sin 4x}{4x} \to 2 \cdot 1 = 2$.
Why not the others: (B) forgets to divide by 2 (gives 4). (C) inverts the factor. (D) uses $\frac{\sin u}{u} \to 1$ without adjusting the constant.

6. Answer: C.
1. $-(x-1)^2 \leq f(x) - 4 \leq (x-1)^2$, so $4 - (x-1)^2 \leq f(x) \leq 4 + (x-1)^2$. 2. Both bounds $\to 4$ as $x \to 1$. 3. By the Squeeze Theorem $\lim\limits_{x\to 1} f(x) = 4$.
Why not the others: (A) mistakes the bound $(x-1)^2 \to 0$ for the limit of f. (B) uses the input value. (D) does not recognise the squeeze.

7. Answer: B.
Same degrees: divide by x^3: $\dfrac{5 - 2/x^2}{2 + 7/x} \to \dfrac{5}{2}$.
Why not the others: (A) assumes the denominator "wins". (C) inverts the leading coefficients. (D) assumes the numerator wins.

8. Answer: D.
1. $f(x) = \dfrac{(x-2)(x+2)}{(x-2)(x-3)}$. 2. The factor $x-2$ cancels (a hole at $x = 2$, with limit $\frac{4}{-1} = -4$). 3. $\dfrac{x+2}{x-3}$ has numerator $5 \neq 0$ at $x = 3$, so $x = 3$ is the only vertical asymptote.
Why not the others: (A) calls the hole an asymptote and misses $x = 3$. (B) uses every zero of the original denominator. (C) uses a zero of the numerator.

9. Answer: A.
1. $\dfrac{x+2}{(x-1)(x+1)}$. As $x \to 1^+$: numerator $\to 3 > 0$, $x + 1 \to 2$, $x - 1 \to 0^+$. 2. So the quotient is positive and grows without bound: ∞.
Why not the others: (B) uses the sign for $x < 1$. (C) treats $\frac{k}{0}$ as 0. (D) substitutes $x = 1$ into the factors that do not vanish and drops the 0.

10. Answer: B.
1. $f(2) = 4 + k$ and $\lim\limits_{x\to 2^-} f = 2k - 1$, $\lim\limits_{x\to 2^+} f = 4 + k$. 2. $2k - 1 = 4 + k \Rightarrow k = 5$.
Why not the others: (A) drops the -1 ($2k = 4 + k$). (C) $2k - 1 = 4 - k$ (sign error on k). (D) $2k + 1 = 4 + k$ (sign error on the 1).

11. Answer: C.
f is continuous on $[1,5]$ and 0 lies between $f(1)=-3$ and $f(5)=8$; the IVT gives c in $(1,5)$ with $f(c)=0$.
Why not the others: (A) the IVT does not force a specific value at 3. (B) continuity does not imply monotonic. (D) 10 is outside $[-3,8]$.

12. Answer: D.
$\dfrac{x^2-1}{x-1}=x+1$ for $x\neq 1$: the limit 2 exists but $f(1)$ is undefined: removable.
Why not the others: (A) infinite discontinuity. (B) jump discontinuity (-1 on the left, $+1$ on the right). (C) is continuous at 1 (no discontinuity).

13. Answer: A.
1. Substitution gives $\frac{0}{0}$. 2. Multiply by the conjugate: $\dfrac{(x+16)-16}{x(\sqrt{x+16}+4)}=\dfrac{1}{\sqrt{x+16}+4}$. 3. Limit $=\dfrac{1}{4+4}=\dfrac{1}{8}$.
Why not the others: (B) $\frac{1}{\sqrt{16}}$, forgetting the $+4$. (C) $\frac{0}{0}=0$. (D) inverts the result.

14. Answer: C.
1. For $x<0$, $\sqrt{9x^2+x}=|x|\sqrt{9+\frac{1}{x}}=-x\sqrt{9+\frac{1}{x}}$. 2. So the quotient is $\dfrac{-x\sqrt{9+1/x}}{x(2+1/x)}=\dfrac{-\sqrt{9+1/x}}{2+1/x}\to\dfrac{-3}{2}$.
Why not the others: (A) uses $\sqrt{x^2}=x$ (valid for $x\to+\infty$ only). (B) forgets the 2 in the denominator. (D) treats the limit at infinity as 0.

Part B solutions

Multiple choice, graphing calculator allowed

15. Answer: B.
1. Table: $x=0.001$ gives $\dfrac{e^{0.003}-1}{0.001}=3.0015$; $x=-0.001$ gives 2.9985. 2. Both sides approach 3. (Algebraically $\frac{e^{3x}-1}{3x}\cdot 3\to 3$.)
Why not the others: (A) forgets the factor 3. (C) inverts. (D) $\frac{0}{0}=0$.

16. Answer: D.
1. $\dfrac{\sin(x-2)}{x^2-4}=\dfrac{\sin(x-2)}{x-2}\cdot\dfrac{1}{x+2}\to 1\cdot\dfrac{1}{4}=0.25$. A calculator table in radian mode confirms it.
Why not the others: (A) degree mode: $\sin(x-2)^\circ\approx\frac{\pi}{180}(x-2)$ gives $\frac{\pi}{720}\approx 0.004$. (B) evaluates $\frac{1}{x+2}$ with $x+2$ replaced by 2. (C) uses only $\frac{\sin u}{u}\to 1$.

17. Answer: A.
g is continuous. $g(0)=1>0$ and $g(1)=e-4\approx -1.282<0$: opposite signs, so IVT gives a solution in $(0,1)$. Other intervals: $g(1)\approx -1.282$, $g(2)\approx -0.611$ (same sign); $g(3)\approx 8.086$, $g(4)\approx 38.6$ (same sign); $g(-1)\approx 4.368$, $g(0)=1$ (same sign).
Why not the others: no sign change on (B), (C), (D), so the IVT guarantees nothing there (note that $g(x)=0$ also has a solution in $(2,3)$, but that interval is not offered).

18. Answer: C.
$\dfrac{f(3)-f(1)}{3-1}=\dfrac{3e^{-3}-e^{-1}}{2}=\dfrac{0.14936-0.36788}{2}=-0.109.$
Why not the others: (A) sign error. (B) -0.219 forgets to divide by 2. (D) divides by 3.

19. Answer: B.
$\dfrac{\tan 3x}{\sin 5x}=\dfrac{\sin 3x}{3x}\cdot\dfrac{5x}{\sin 5x}\cdot\dfrac{3}{5}\cdot\dfrac{1}{\cos 3x}\to\dfrac{3}{5}=0.6.$
Why not the others: (A) inverts the ratio. (C) ignores the coefficients. (D) $\frac{0}{0}=0$.

20. Answer: D.
As $x\to\infty$: $\dfrac{2x-1}{|x|\sqrt{1+4/x^2}}\to\dfrac{2x}{x}=2$. As $x\to-\infty$, $|x|=-x$, so the limit is -2.
Why not the others: (A) and (B) look at only one end. (C) treats $\sqrt{x^2+4}$ as having higher degree than $2x-1$.

21. Answer: A.
Sign changes of g across the table: $g(0) < 0 < g(4)$ gives a zero in $(0,4)$; $g(4) > 0 > g(7)$ gives one in $(4,7)$; $g(7) < 0 < g(10)$ gives one in $(7,10)$. These intervals do not overlap: at least 3 zeros.
Why not the others: (B) misses the interval $(7,10)$. (C) counts only one sign change. (D) counts table entries instead of sign changes.

Part C solutions

Free response, scoring-guide style

Question 22 (9 points).
(a) $\lim_{t\to 1^-} C(t) = 40(1) = 40.$ **1 point**
$\lim_{t\to 1^+} C(t) = 100 - k\,e^{0} = 100 - k.$ **1 point**
(b) $C(1) = 40$ (the first piece includes $t = 1$). Continuity requires $100 - k = 40$, so $k = 60$. **1 point**
Justification: with $k = 60$, $C(1) = 40$, $\lim_{t\to 1^-} C(t) = 40$ and $\lim_{t\to 1^+} C(t) = 100 - 60 = 40$, so $\lim_{t\to 1} C(t) = 40 = C(1)$. **1 point**
(c) As $t \to \infty$, $e^{-0.5(t-1)} \to 0$, so $\lim_{t\to\infty} C(t) = 100$. **1 point**
Interpretation: in the long run the charge level approaches 100 percent (a full charge). **1 point**
(d) For $t \geq 2$, $C(t) = 100 - 60e^{-0.5(t-1)}$ is continuous (composition of continuous functions), so C is continuous on $[2,5]$. $C(2) = 100 - 60e^{-0.5} = 63.608$ and $C(5) = 100 - 60e^{-2} = 91.880$. **1 point**
Since $63.608 < 80 < 91.880$, by the Intermediate Value Theorem there is t in $(2,5)$ with $C(t) = 80$. **1 point**
Solve $100 - 60e^{-0.5(t-1)} = 80$: $e^{-0.5(t-1)} = \dfrac{1}{3}$, $-0.5(t-1) = -\ln 3$, $t = 1 + 2\ln 3 = 3.197$ hours. **1 point**

Question 23 (9 points).
(a) $g(x) = \dfrac{(x-4)(x+2)}{(x-4)(x+4)} = \dfrac{x+2}{x+4}$ for $x \neq \pm 4$. **1 point**
$\lim_{x\to 4} g(x) = \dfrac{4+2}{4+4} = \dfrac{3}{4}.$ **1 point**
(b) The factor $x - 4$ cancels (a hole at $x = 4$), so the vertical asymptote is at $x = -4$ only. **1 point**
For x near -4 the numerator $x + 2 \to -2 < 0$. As $x \to -4^-$, $x + 4 \to 0^-$, so $\dfrac{x+2}{x+4} \to +\infty$. **1 point**
As $x \to -4^+$, $x + 4 \to 0^+$, so $\dfrac{x+2}{x+4} \to -\infty$. **1 point**
(c) h is continuous at 4 iff $\lim_{x\to 4} h(x) = h(4)$. By part (a) the limit is $\dfrac{3}{4}$, so $k = \dfrac{3}{4}$. **1 point**
Justification: with $k = \frac{3}{4}$, $h(4) = \frac{3}{4}$ exists, $\lim_{x\to 4} h(x) = \dfrac{3}{4}$ exists, and they are equal. **1 point**
(d) $\lim_{x\to\infty} g(x) = \lim_{x\to\infty} \dfrac{x+2}{x+4} = \lim_{x\to\infty} \dfrac{1 + 2/x}{1 + 4/x} = 1.$ **1 point**
The horizontal asymptote is $y = 1$. **1 point**

Unit 2 Solutions

Check-yourself answers

Topics 2.1–2.10

Topic 2.1 check.

1. $\dfrac{f(4)-f(1)}{4-1}=\dfrac{4-(-2)}{3}=2.$

2. $f'(2)=\lim\limits_{h\to0}\dfrac{(2+h)^2-3(2+h)-(-2)}{h}=\lim\limits_{h\to0}\dfrac{h+h^2}{h}=\lim\limits_{h\to0}(1+h)=1.$

3. Average velocity $=\dfrac{s(3)-s(1)}{3-1}=\dfrac{9-(-5)}{2}=7$ m/s. Instantaneous: $v(2)=\lim\limits_{h\to0}\dfrac{s(2+h)-s(2)}{h}=\lim\limits_{h\to0}\dfrac{(2+h)^3-6(2+h)+4}{h}=\lim\limits_{h\to0}\dfrac{6h+6h^2+h^3}{h}=\lim\limits_{h\to0}(6+6h+h^2)=6$ m/s.

Topic 2.2 check.

1. $\dfrac{3(x+h)^2-(x+h)-3x^2+x}{h}=\dfrac{6xh+3h^2-h}{h}=6x+3h-1\to6x-1.$

2. $\dfrac{\frac{1}{x+h}-\frac{1}{x}}{h}=\dfrac{x-(x+h)}{hx(x+h)}=\dfrac{-1}{x(x+h)}\to-\dfrac{1}{x^2}$. At $x=2$: $f(2)=\frac12$, $f'(2)=-\frac14$, so $y-\frac12=-\frac14(x-2)$.

3. $f'(2)=4\cdot2^3=32$ (by the definition, $\lim_{x\to2}\frac{x^4-16}{x-2}=f'(2)$ for $f(x)=x^4$).

Topic 2.3 check.

1. Closest entries around 4: $x=3$ and $x=6$. $f'(4)\approx\dfrac{4-10}{6-3}=-2.$

2. Closest entries around 8: $x=6$ and $x=10$. $f'(8)\approx\dfrac{-6-4}{10-6}=-2.5.$

3. $\dfrac{2.1^3-1.9^3}{0.2}=\dfrac{9.261-6.859}{0.2}=12.01$; exact $f'(2)=3(2)^2=12$. The estimate is high by 0.01.

Topic 2.4 check.

1. Right-hand slope: $\frac{|h|}{h}=1$ for $h>0$; left-hand slope: $\frac{|h|}{h}=-1$ for $h<0$. Unequal, so f is **not** differentiable at $x=3$.

2. Continuity: $1=a+b$. Slopes: $2(1)=a$. So $a=2$, $b=-1$.

3. $h(0)=0=\lim_{x\to0}x^{2/3}$, so h is continuous at 0. $\dfrac{h(0+t)-h(0)}{t}=t^{-1/3}$, which tends to $+\infty$ as $t\to0^+$ and to $-\infty$ as $t\to0^-$. The limit does not exist, so h is not differentiable at 0 (cusp).

Topic 2.5 check.

1. $f'=7x^6$; $g'=-3x^{-4}$; $h=x^{2/3}$, $h'=\frac23x^{-1/3}$.

2. $f(x)=x^{-3/2}$, $f'(x)=-\frac32x^{-5/2}$, $f'(4)=-\frac32\cdot\frac{1}{32}=-\dfrac{3}{64}.$

3. $4x^3=32\Rightarrow x^3=8\Rightarrow x=2.$

Topic 2.6 check.

1. $f'(x)=12x^2-10x+7$, so $f'(1)=9$.

2. $y=3x^{1/2}-6x^{-2}+\pi^2$, $\dfrac{dy}{dx}=\frac32x^{-1/2}+12x^{-3}$. At $x=4$: $\frac32\cdot\frac12+\frac{12}{64}=\frac34+\frac{3}{16}=\dfrac{15}{16}.$

3. $f'(x)=3x^2-6x-9=3(x-3)(x+1)=0\Rightarrow x=3$ or $x=-1$.

Topic 2.7 check.

1. $f'(x)=3\cos x+2\sin x+5e^x$; $f'(0)=3+0+5=8.$

2. $g'(x)=\frac{4}{x}-e^x$; $g'(1)=4-e.$

3. At $x=\frac{\pi}{2}$: $y=0$, slope $-\sin\frac{\pi}{2}=-1$. Tangent: $y=-\left(x-\frac{\pi}{2}\right)$.

Topic 2.8 check.

1. $2xe^x+x^2e^x=(x^2+2x)e^x$; at $x=1$: $3e$.

2. $\ln x+x\cdot\frac{1}{x}=\ln x+1$; at $x=e$: 2.

3. $h'(2)=f'(2)g(2)+f(2)g'(2)=(-1)(4)+(3)(5)=11.$

Topic 2.9 check.

1. $y' = \dfrac{2(x+3)-(2x-1)(1)}{(x+3)^2} = \dfrac{7}{(x+3)^2}$.

2. $\dfrac{e^x x^2 - e^x(2x)}{x^4} = \dfrac{e^x(x-2)}{x^3}$; at $x=2$: 0.

3. $h'(1) = \dfrac{f'(1)g(1) - f(1)g'(1)}{g(1)^2} = \dfrac{2(3)-6(-1)}{9} = \dfrac{12}{9} = \dfrac{4}{3}$.

Topic 2.10 check.

1. $\sec^2 x - 2\sec x \tan x$. At $\frac{\pi}{4}$: $\sec^2 = 2$, $\sec = \sqrt{2}$, $\tan = 1$: $2 - 2\sqrt{2}$.

2. $\cot x + x(-\csc^2 x) = \cot x - x\csc^2 x$.

3. $\dfrac{d}{dx}\sec x = \sec x \tan x$; at $\frac{\pi}{3}$: $2 \cdot \sqrt{3} = 2\sqrt{3}$.

Part A – Multiple choice, no calculator

Answers and reasoning

1. Answer: B.
1. The limit is the definition of $f'(2)$ for $f(x) = x^3$ (since $f(2) = 8$). 2. $f'(x) = 3x^2$, so $f'(2) = 12$.
Why not the others: (A) substituting $h = 0$ and calling $\frac{0}{0}$ "0"; (C) differentiating x^3 as $3x$; (D) reporting $f(2) = 8$ instead of the slope.

2. Answer: C.
1. $f(2) = 8 - 2 = 6$, $f(0) = 0$. 2. $\dfrac{6-0}{2-0} = 3$.
Why not the others: (A) $f'(1) = 2$, the slope at the midpoint (instantaneous, not average); (B) forgetting to divide by $b - a = 2$; (D) $f'(2) = 11$, the instantaneous rate at the endpoint.

3. Answer: A.
1. $f(x) = x^3 - 8x^{1/2}$, $f'(x) = 3x^2 - 4x^{-1/2}$. 2. $f'(4) = 3(16) - \dfrac{4}{2} = 48 - 2 = 46$.
Why not the others: (B) $\frac{d}{dx}\sqrt{x}$ written as $\frac{1}{2}\sqrt{x}$ (exponent not reduced), giving $48 - 8 = 40$; (C) sign error on the second term, $48 + 2$; (D) dropping the $\frac{1}{2}$, $\frac{d}{dx}(8\sqrt{x}) = 8x^{-1/2} = 4$, giving 44.

4. Answer: D.
1. $\dfrac{d}{dx}[x^3 \ln x] = 3x^2 \ln x + x^3 \cdot \dfrac{1}{x} = 3x^2 \ln x + x^2$. 2. At $x = e$: $3e^2(1) + e^2 = 4e^2$.
Why not the others: (A) product of derivatives $3x^2 \cdot \frac{1}{x} = 3e$; (B) only the second term e^2; (C) only the first term $3e^2$.

5. Answer: C.
1. $f'(x) = \dfrac{(1)(x-1)-(x+1)(1)}{(x-1)^2} = \dfrac{-2}{(x-1)^2}$. 2. $f'(3) = \dfrac{-2}{4} = -\dfrac{1}{2}$.
Why not the others: (A) quotient of derivatives $\frac{1}{1}$; (B) forgetting to square the denominator, $\frac{-2}{2}$; (D) numerator subtracted in the wrong order, $+2$.

6. Answer: D.
1. The table entries closest to $x = 5$ are $x = 4$ and $x = 6$. 2. $g'(5) \approx \dfrac{16-10}{6-4} = 3$.
Why not the others: (A) using the endpoints 0 and 10: $\frac{20-3}{10} = 1.7$; (B) not dividing by $6 - 4$; (C) pairing $x = 0$ with $x = 6$: $\frac{16-3}{6}$.

7. Answer: A.
1. Continuity at $x = 1$: $1^3 = a + b$, so $a + b = 1$. 2. Equal slopes: $3(1)^2 = 2a(1) + b$, so $2a + b = 3$. 3. Subtracting, $a = 2$ and $b = -1$.
Why not the others: (B) taking the right-hand derivative as a (forgetting the power x^2), then $b = 1 - 3$; (C) satisfies continuity only, slope $2 \neq 3$; (D) setting both equations equal to 3, forgetting that the function value at 1 is 1.

8. Answer: B.
1. $x = 2$: left slope 1, right slope -1 (a corner). 2. $x = 4$: $\lim_{x \to 4^-} f(x) = 1 \neq f(4) = 2$ (a discontinuity, so not

differentiable). 3. $x = 5$: left slope 0, right slope 2 (a corner). The graph is smooth elsewhere on $(0, 6)$: 3 points.
Why not the others: (A) counting only the discontinuity; (C) counting only the two corners and forgetting that a discontinuity prevents differentiability; (D) also counting an endpoint or another point where the graph is smooth.

9. Answer: C.
1. $\dfrac{d}{dx}[2\sec x - \tan x] = 2\sec x \tan x - \sec^2 x$. 2. At $\frac{\pi}{3}$: $\sec = 2$, $\tan = \sqrt{3}$: $2(2)(\sqrt{3}) - 4 = 4\sqrt{3} - 4$.
Why not the others: (A) sign error on $\sec^2 x$; (B) using $\frac{d}{dx}\sec x = \sec^2 x$, giving $8 - 4$; (D) losing the factor 2.

10. Answer: A.
1. $y' = \dfrac{e^x \cdot x - e^x \cdot 1}{x^2} = \dfrac{e^x(x-1)}{x^2}$. 2. At $x = 2$: $\dfrac{e^2}{4}$.
Why not the others: (B) a plus sign in the numerator, $\frac{2e^2+e^2}{4}$; (C) forgetting to square the denominator, $\frac{e^2}{2}$; (D) quotient of derivatives, $\frac{e^2}{1}$.

11. Answer: B.
1. $f'(x) = 2x\ln x + x^2 \cdot \frac{1}{x} = x(2\ln x + 1)$. 2. For $x > 0$, $f'(x) = 0$ when $2\ln x + 1 = 0$, so $\ln x = -\frac{1}{2}$ and $x = e^{-1/2}$.
Why not the others: (A) the second term lost, $2x\ln x = 0$ at $x = 1$; (C) sign slip, $\ln x = +\frac{1}{2}$; (D) $\ln x = -2$, forgetting to divide by 2.

12. Answer: C.
1. $k'(x) = 2xf(x) + x^2 f'(x)$. 2. $k'(3) = 2(3)(-2) + 9(5) = -12 + 45 = 33$.
Why not the others: (A) product of derivatives $2x \cdot f'(3) = 30$; (B) only the second term $9 \cdot 5$; (D) using $|f(3)|$, $+12 + 45$.

13. Answer: D.
1. With $f(x) = \sin x$, $a = \frac{\pi}{6}$, $f(a) = \frac{1}{2}$, the limit is $f'\left(\frac{\pi}{6}\right)$. 2. $f'(x) = \cos x$, so the value is $\cos\frac{\pi}{6} = \dfrac{\sqrt{3}}{2}$.
Why not the others: (A) "numerator and denominator both $\to 0$", so 0; (B) giving $f(a) = \sin\frac{\pi}{6}$ instead of the slope; (C) using $\frac{d}{dx}\sin x = -\cos x$.

14. Answer: B.
1. Differentiability at 3 implies continuity at 3 (I true). 2. Continuity means $\lim_{x\to 3} f(x) = f(3)$ exists (II true). 3. Differentiable does not imply that f' is continuous: e.g. $f(x) = x^2 \sin\frac{1}{x}$ ($f(0) = 0$) is differentiable at 0 but f' is not continuous there (III need not be true).
Why not the others: (A) forgetting that continuity includes the limit existing; (C), (D) assuming the derivative itself must be continuous.

Part B – Multiple choice, calculator allowed

Answers and reasoning

15. Answer: C.
1. $f'(x) = e^x \ln x + e^x \cdot \frac{1}{x}$. 2. $f'(2) = e^2 \ln 2 + \dfrac{e^2}{2} \approx 5.122 + 3.695 = 8.816$.
Why not the others: (A) minus sign between the terms, $5.122 - 3.695$; (B) product of derivatives, $e^2 \cdot \frac{1}{2}$; (D) only the first term, $e^2 \ln 2$.

16. Answer: A.
1. The smallest interval of the table containing $t = 7$ is $[5, 9]$. 2. $H'(7) \approx \dfrac{57 - 71}{9 - 5} = -3.5$ degrees Celsius per minute.
Why not the others: (B) using $t = 2$ and $t = 9$, $\frac{57-88}{7}$; (C) using the interval $[2, 5]$, which does not contain 7; (D) subtracting in the wrong order, losing the sign.

17. Answer: D.
1. $y' = 2x + e^x$. 2. Solve $2x + e^x = 5$ on the calculator (intersection of $y = 2x + e^x$ and $y = 5$): $x \approx 1.059$.
Why not the others: (A) solving $y = 5$ (height 5) instead of $y' = 5$ (slope 5); (B) ignoring the $2x$ term, $e^x = 5$; (C) ignoring the e^x term, $2x = 5$.

18. Answer: B.
1. $f'(x) = \tan x + x\sec^2 x$. 2. With the calculator in radian mode, $f'(2) = \tan 2 + 2\sec^2 2 \approx -2.185 + 11.549 = 9.364$.
Why not the others: (A) calculator in degree mode; (C) forgetting the factor x on $\sec^2 x$, $\tan 2 + \sec^2 2$; (D) only $\sec^2 2$, i.e. $\frac{d}{dx}\tan x$ without the Product Rule.

19. Answer: C.
1. $g'(x) = \dfrac{2x\ln x - x^2 \cdot \frac{1}{x}}{(\ln x)^2} = \dfrac{x(2\ln x - 1)}{(\ln x)^2}$. 2. $g'(3) = \dfrac{3(2\ln 3 - 1)}{(\ln 3)^2} \approx 2.976$.
Why not the others: (A) numerator subtracted in the wrong order; (B) forgetting to square $\ln x$, $\frac{3(2\ln 3-1)}{\ln 3}$; (D) quotient of derivatives, $\frac{2x}{1/x} = 2x^2$.

20. Answer: A.
1. $f(2) = 8\ln 2 \approx 5.545$. 2. $f'(x) = 3x^2\ln x + x^2$, so $f'(2) = 12\ln 2 + 4 \approx 12.318$. 3. Tangent: $y = f(2) + f'(2)(x-2)$; at $x = 0$: $y = 5.545 - 2(12.318) = -19.090$.
Why not the others: (B) omitting $f(2)$; (C) reporting $f(2)$; (D) reporting the slope $f'(2)$.

21. Answer: D.
1. $f'(x) = 1 + 2\cos x$. 2. $f'(x) = 0 \iff \cos x = -\frac{1}{2}$. In $(0, 10)$ the solutions are $x = \frac{2\pi}{3} \approx 2.094$, $\frac{4\pi}{3} \approx 4.189$, $\frac{8\pi}{3} \approx 8.378$; the next, $\frac{10\pi}{3} \approx 10.47$, is outside. So there are 3 points.
Why not the others: (A) stopping after the first period; (B) including $\frac{10\pi}{3}$, which exceeds 10; (C) solving $\cos x = \pm\frac{1}{2}$.

Part C – Free response

Scoring guide style

22. (Calculator active, 9 points)

(a) *2 points.*
$\dfrac{W(4) - W(1)}{4 - 1} = \dfrac{(12 + 16\ln 4) - 3}{3} \approx \dfrac{34.181 - 3}{3} = 10.394.$
1 point: correct difference quotient with the values $W(4)$, $W(1)$. *1 point:* answer 10.394 thousand litres per hour (units required).

(b) *3 points.*
$W'(t) = 3 + \left(2t\ln t + t^2 \cdot \frac{1}{t}\right) = 3 + 2t\ln t + t$. *1 point:* Product Rule term $2t\ln t + t$. *1 point:* the rest of $W'(t)$ correct.
$W'(2) = 3 + 4\ln 2 + 2 \approx 7.773 > 0$. *1 point:* "At $t = 2$ hours the amount of water in the reservoir is increasing at a rate of 7.773 thousand litres per hour" (units, context, and "increasing" because $W'(2) > 0$).

(c) *2 points.*
$W(2) = 6 + 4\ln 2 \approx 8.773$ and $W'(2) \approx 7.773$. *1 point:* slope. *1 point:* equation $y - 8.773 = 7.773(t - 2)$, that is $W \approx 8.773 + 7.773(t - 2)$.

(d) *2 points.*
The smallest table interval containing $t = 3$ is $[2.5, 4]$: $V'(3) \approx \dfrac{V(4) - V(2.5)}{4 - 2.5} = \dfrac{25 - 21}{1.5} \approx 2.667$ thousand litres per hour. *1 point:* difference quotient with correct entries. *1 point:* value with units.

23. (No calculator, 9 points)

(a) *2 points.*
$h'(x) = f'(x)g(x) + f(x)g'(x)$ *(1 point, Product Rule).* $h'(2) = 6 \cdot 2 + 5 \cdot 3 = 12 + 15 = 27$ *(1 point).*

(b) *2 points.*
$p'(3) = \dfrac{f'(3)g(3) - f(3)g'(3)}{[g(3)]^2} = \dfrac{2 \cdot 4 - 1 \cdot (-1)}{16} = \dfrac{9}{16}$ *(1 point).* $p(3) = \dfrac{1}{4}$. Tangent line: $y - \dfrac{1}{4} = \dfrac{9}{16}(x - 3)$ *(1 point).*

(c) *3 points.*
Differentiable implies continuous, so the pieces agree at $x = 2$: $f(2) = a\,g(2) + b$, that is $5 = 2a + b$ *(1 point).* The one-sided derivatives must be equal: $f'(2) = a\,g'(2)$, that is $6 = 3a$ *(1 point).* Therefore $a = 2$ and $b = 5 - 4 = 1$ *(1 point).*

(d) *2 points.*
$w'(x) = f'(x) - 2$, so $w'(3) = 2 - 2 = 0$: the tangent line is horizontal *(1 point).* $w(3) = f(3) - 6 = 1 - 6 = -5$,

so the line is $y = -5$ *(1 point)*.

Unit 3 Solutions

Topic check answers

Topic 3.1 check

1. Outer $\cos u$, inner $u = x^3$: $\dfrac{dy}{dx} = -\sin(x^3) \cdot 3x^2 = -3x^2 \sin(x^3)$.

2. $f(x) = (4 - x^2)^{1/2}$, so $f'(x) = \dfrac{1}{2}(4 - x^2)^{-1/2}(-2x) = -\dfrac{x}{\sqrt{4 - x^2}}$. f is defined for $-2 \le x \le 2$; f' is defined for $-2 < x < 2$ (the denominator is 0 at ± 2).

3. $h'(3) = f'(g(3))\, g'(3) = f'(2) \cdot 7 = (-2)(7) = -14$.

Topic 3.2 check

1. Differentiate: $e^y \dfrac{dy}{dx} = 1 + \dfrac{dy}{dx}$. Then $(e^y - 1)\dfrac{dy}{dx} = 1$, so $\dfrac{dy}{dx} = \dfrac{1}{e^y - 1}$.

2. The point is on the curve: $1 - 2 + 4 = 3$. Differentiating: $2x - y - x\dfrac{dy}{dx} + 2y\dfrac{dy}{dx} = 0$, so $\dfrac{dy}{dx} = \dfrac{y - 2x}{2y - x}$. At $(1, 2)$: $\dfrac{2 - 2}{4 - 1} = 0$. The tangent line is $y = 2$.

3. $1 \cdot 1 + \ln 1 = 1$, so $(1, 1)$ is on the curve. Differentiating: $y + x\dfrac{dy}{dx} + \dfrac{1}{y}\dfrac{dy}{dx} = 0$. At $(1, 1)$: $1 + \dfrac{dy}{dx} + \dfrac{dy}{dx} = 0$, so $\dfrac{dy}{dx} = -\dfrac{1}{2}$.

Topic 3.3 check

1. $f(1) = 3$, so $g(3) = 1$. $f'(x) = 5x^4 + 3x^2 + 1$, $f'(1) = 9$. Hence $g'(3) = \dfrac{1}{f'(1)} = \dfrac{1}{9}$.

2. $f(2) = 5$ gives $f^{-1}(5) = 2$, and $(f^{-1})'(5) = \dfrac{1}{f'(2)} = \dfrac{1}{3}$. Tangent line: $y - 2 = \dfrac{1}{3}(x - 5)$.

3. For $f(x) = x^3$, $g(x) = \sqrt[3]{x}$ and $g(0) = 0$. Since $f'(g(0)) = f'(0) = 0$, the formula $g'(0) = \dfrac{1}{f'(0)}$ would need division by zero: the graph of g has a vertical tangent at the origin, so $g'(0)$ does not exist.

Topic 3.4 check

1. $u = e^x$: $\dfrac{e^x}{1 + (e^x)^2} = \dfrac{e^x}{1 + e^{2x}}$.

2. $f'(x) = \dfrac{1/2}{\sqrt{1 - x^2/4}}$, so $f'(1) = \dfrac{1/2}{\sqrt{3}/2} = \dfrac{1}{\sqrt{3}}$.

3. $\dfrac{dy}{dx} = -\dfrac{2}{\sqrt{1 - 4x^2}}$; at $x = 0$ the value is -2.

Topic 3.5 check

1. Product rule: $\dfrac{dy}{dx} = 3x^2 \ln(2x) + x^3 \cdot \dfrac{2}{2x} = 3x^2 \ln(2x) + x^2$.

2. Quotient rule: numerator $= (-\sin x)(1 + \sin x) - \cos x \cdot \cos x = -\sin x - \sin^2 x - \cos^2 x = -\sin x - 1$. So the derivative is $\dfrac{-(1 + \sin x)}{(1 + \sin x)^2} = -\dfrac{1}{1 + \sin x}$.

3. $\dfrac{dy}{dx} = 3^{x^2} \ln 3 \cdot 2x = 2x \ln 3 \cdot 3^{x^2}$.

Topic 3.6 check

1. $f'(x) = 5x^4 - 6x$, $f''(x) = 20x^3 - 6$, $f'''(x) = 60x^2$.

2. $f'(x) = \dfrac{2x}{x^2 + 1}$; $f''(x) = \dfrac{2(x^2 + 1) - 2x \cdot 2x}{(x^2 + 1)^2} = \dfrac{2 - 2x^2}{(x^2 + 1)^2}$. So $f''(1) = 0$.

3. $y + x\dfrac{dy}{dx} = 0$, so $\dfrac{dy}{dx} = -\dfrac{y}{x}$. Then $\dfrac{d^2y}{dx^2} = -\dfrac{\dfrac{dy}{dx} \cdot x - y}{x^2}$. Substituting $\dfrac{dy}{dx} = -\dfrac{y}{x}$: $-\dfrac{-y - y}{x^2} = \dfrac{2y}{x^2}$. At $(1, 1)$: 2. (Check: $y = \dfrac{1}{x}$ gives $y'' = \dfrac{2}{x^3} = 2$.)

Part A: Multiple Choice – No calculator

Answers and reasoning

1. Answer: B. Outer $\sin u$, inner $u = 5x^2$. $f'(x) = \cos(5x^2) \cdot 10x$.
Why not the others: (A) drops the inner derivative $10x$. (C) uses $-\cos$, confusing the sine and cosine derivatives. (D) rewrites $5x^2$ as $10x$ inside the cosine, differentiating the inside *in place*.

2. Answer: D. $h'(2) = f'(g(2))\,g'(2) = f'(3) \cdot g'(2) = 7 \cdot (-4) = -28$.
Why not the others: (A) $7 = f'(g(2))$ only, with the inner derivative omitted. (B) $-24 = f'(2)\,g'(2)$: the outer derivative is evaluated at 2 instead of $g(2) = 3$. (C) $21 = f'(3)\,g'(3)$: the inner derivative is evaluated at the wrong place.

3. Answer: A. $2x + y + x\dfrac{dy}{dx} = 0$ (product rule on xy), so $x\dfrac{dy}{dx} = -(2x + y)$ and $\dfrac{dy}{dx} = -\dfrac{2x + y}{x}$.
Why not the others: (B) -2 forgets the term y from the product rule ($2x + x\dfrac{dy}{dx} = 0$). (C) fails to divide by x. (D) has a sign error when moving $2x + y$ to the right.

4. Answer: C. $f(1) = 2 + 3 = 5$, so $g(5) = 1$. $f'(x) = 6x^2 + 3$, $f'(1) = 9$. $g'(5) = \dfrac{1}{f'(g(5))} = \dfrac{1}{9}$.
Why not the others: (A) $\dfrac{1}{5}$ takes the reciprocal of the input. (B) $9 = f'(g(5))$: the reciprocal is forgotten. (D) $\dfrac{1}{153} = \dfrac{1}{f'(5)}$: f' is evaluated at 5 rather than at $g(5) = 1$.

5. Answer: B. $\dfrac{d}{dx}\arctan(x^2) = \dfrac{2x}{1 + (x^2)^2} = \dfrac{2x}{1 + x^4}$; at $x = 1$: $\dfrac{2}{2} = 1$.
Why not the others: (A) $\dfrac{1}{2} = \dfrac{1}{1 + x^4}$ at 1: the inner derivative $2x$ is missing. (C) $2 = 2x$ only: the factor $\dfrac{1}{1 + u^2}$ is missing. (D) $4 = 2x(1 + x^4)$: the arctangent factor is multiplied instead of divided.

6. Answer: A. $f'(x) = 2xe^x + x^2e^x = (x^2 + 2x)e^x$. $f''(x) = (2x + 2)e^x + (x^2 + 2x)e^x = (x^2 + 4x + 2)e^x$. $f''(1) = 7e$.
Why not the others: (B) $3e = f'(1)$ (only one derivative taken). (C) $4e$ keeps only the cross term $2 \cdot 2x\,e^x$ of the product rule for second derivatives. (D) 7 evaluates the polynomial $x^2 + 4x + 2$ and loses the factor e^x.

7. Answer: D. $h'(x) = f'(x^2) \cdot 2x$, so $h'(2) = f'(4) \cdot 4 = (-2)(4) = -8$.
Why not the others: (A) $-2 = f'(4)$ only, inner derivative omitted. (B) $12 = f'(2) \cdot 4$: f' is evaluated at $x = 2$ instead of $x^2 = 4$. (C) $1 = f'(2) + f'(4)$: the derivatives are added instead of using the chain rule.

8. Answer: C. $3x^2 + 3y^2\dfrac{dy}{dx} = 0$, so $\dfrac{dy}{dx} = -\dfrac{x^2}{y^2}$. At $(1, 2)$: $-\dfrac{1}{4}$. Line: $y - 2 = -\dfrac{1}{4}(x - 1)$.
Why not the others: (A) inverts the fraction: $-\dfrac{y^2}{x^2} = -4$. (B) drops the minus sign. (D) $-\dfrac{1}{2} = -\dfrac{x^2}{y}$: the exponent on y in the denominator is lost.

9. Answer: B. $f'(x) = \dfrac{1}{\sqrt{1 - (\sqrt{x})^2}} \cdot \dfrac{1}{2\sqrt{x}} = \dfrac{1}{\sqrt{1 - x}} \cdot \dfrac{1}{2\sqrt{x}}$. At $x = \dfrac{1}{2}$: $\sqrt{2} \cdot \dfrac{1}{2/\sqrt{2}} = \sqrt{2} \cdot \dfrac{\sqrt{2}}{2} = 1$.
Why not the others: (A) $\dfrac{1}{\sqrt{2}} = \dfrac{1}{2\sqrt{x}}$ is the inner derivative alone: the arcsine factor is missing. (C) $\sqrt{2} = \dfrac{1}{\sqrt{1 - x}}$ only: the inner derivative is missing. (D) $2 = \dfrac{1}{1 - x}$: the radical is dropped and the inner derivative is missing.

10. Answer: A. Quotient rule: numerator $-\sin x(1 + \sin x) - \cos^2 x = -(1 + \sin x)$, so the derivative is $-\dfrac{1}{1 + \sin x}$; at $\dfrac{\pi}{2}$ this is $-\dfrac{1}{2}$.
Why not the others: (B) -2 inverts the result. (C) $\dfrac{1}{2}$ has a sign error in $(\cos x)'$. (D) 0 keeps only the term

$-uv' = -\cos^2 x$ (which is 0 at $\pi/2$) and drops $u'v$.

11. Answer: D. $\dfrac{dy}{dx} = -\dfrac{x}{y}$; $\dfrac{d^2y}{dx^2} = -\dfrac{y - x\dfrac{dy}{dx}}{y^2} = -\dfrac{y + x^2/y}{y^2} = -\dfrac{x^2+y^2}{y^3} = -\dfrac{25}{y^3}$. At $(3,4)$: $-\dfrac{25}{64}$.
Why not the others: (A) $-\dfrac{3}{4}$ is $\dfrac{dy}{dx}$, the first derivative. (B) loses the minus sign. (C) $-\dfrac{1}{4} = -\dfrac{y^2}{y^3}$: the x^2 term is dropped.

12. Answer: C. The tangent line through $(2,5)$ and $(3,1)$ has slope $\dfrac{1-5}{3-2} = -4$, so $f'(2) = -4$ and $g(5) = 2$. Then $g'(5) = \dfrac{1}{f'(2)} = -\dfrac{1}{4}$, and the tangent line to g passes through $(5,2)$: $y - 2 = -\dfrac{1}{4}(x-5)$.
Why not the others: (A) swaps the coordinates but keeps the slope -4. (B) keeps the point $(2,5)$ of f and only reciprocates the slope. (D) uses $+\dfrac{1}{4}$: the sign of the reciprocal is lost.

13. Answer: A. $f'(x) = e^{\arctan x} \cdot \dfrac{1}{1+x^2}$; at $x = 1$: $\dfrac{e^{\pi/4}}{2}$.
Why not the others: (B) omits the factor $\dfrac{1}{1+x^2}$. (C) uses $\dfrac{1}{\sqrt{1+x^2}}$, a radical that does not belong to the arctangent derivative. (D) multiplies by $1+x^2$ instead of dividing.

14. Answer: C. $2x - 2y - 2x\dfrac{dy}{dx} + 4y\dfrac{dy}{dx} = 0$, so $\dfrac{dy}{dx} = \dfrac{y-x}{2y-x}$. A horizontal tangent needs $y = x$ (and $2y - x = x \neq 0$). Substituting: $x^2 - 2x^2 + 2x^2 = x^2 = 5$, so $a = \sqrt{5}$ (positive).
Why not the others: (A) 1 comes from $y = -x$ (a sign error in the numerator, giving $5x^2 = 5$). (B) $\sqrt{10}$ sets the *denominator* to zero ($y = \dfrac{x}{2}$), which finds the vertical tangents. (D) $5 = a^2$: the square root is not taken.

Part B: Multiple Choice – Graphing calculator allowed

Answers and reasoning

15. Answer: D. $f'(x) = \cos(e^x) \cdot e^x$. At $x = 1.2$: $e^{1.2} \approx 3.3201$, $\cos(3.3201) \approx -0.9841$, so $f'(1.2) \approx -3.267$.
Why not the others: (A) $-0.984 = \cos(e^{1.2})$: the inner derivative e^x is missing. (B) $0.590 = -\sin(e^{1.2})\,e^{1.2}$: differentiates as if the outer function were cosine. (C) $3.315 \approx e^{1.2}$: the outer derivative $\cos(e^x)$ is missing.

16. Answer: A. Differentiate: $\dfrac{dy}{dx} + e^y\dfrac{dy}{dx} = 2x$, so $\dfrac{dy}{dx} = \dfrac{2x}{1+e^y}$. At $x = 2$: $y + e^y = 6$; the calculator gives $y \approx 1.5033$ ($y > 0$), $e^y \approx 4.4967$. Then $\dfrac{dy}{dx} = \dfrac{4}{5.4967} \approx 0.728$.
Why not the others: (B) $0.890 = \dfrac{4}{e^y}$: the term $\dfrac{dy}{dx}$ from the y on the left is lost. (C) $1.598 = \dfrac{4}{1+y}$: e^y is differentiated as if it were y. (D) $1.144 = \dfrac{4}{e^y - 1}$: a sign error when collecting $\dfrac{dy}{dx}$ terms.

17. Answer: C. Solve $2x + e^x = 6$ on the calculator: $g(6) \approx 1.2518$. $f'(x) = 2 + e^x$, so $f'(g(6)) \approx 5.4967$ and $g'(6) = \dfrac{1}{5.4967} \approx 0.182$.
Why not the others: (A) $0.002 = \dfrac{1}{f'(6)}$: f' is evaluated at 6 instead of $g(6)$. (B) $5.496 = f'(g(6))$: the reciprocal is forgotten. (D) $0.212 = \dfrac{1}{f'(1)}$: $g(6)$ is guessed as 1 instead of being solved for.

18. Answer: B. $\theta(t) = \arctan\dfrac{3t^2}{40}$, so $\theta'(t) = \dfrac{1}{1 + \left(\frac{3t^2}{40}\right)^2} \cdot \dfrac{6t}{40}$. At $t = 4$: $u = \dfrac{48}{40} = 1.2$, $\theta'(4) = \dfrac{0.6}{1+1.44} \approx 0.246$ rad/s.
Why not the others: (A) $0.600 = \dfrac{h'(4)}{40}$: the arctangent derivative is omitted. (C) $0.273 = \dfrac{0.6}{1+1.2}$: u is not squared.

(D) $9.836 = \dfrac{24}{1+1.44}$: the division by 40 in u' is forgotten.

19. Answer: D. $s'(t) = \left(2t - \dfrac{t^2}{2}\right)e^{-t/2}$ and $s''(t) = \left(\dfrac{t^2}{4} - 2t + 2\right)e^{-t/2}$. At $t = 3$: $(2.25 - 6 + 2)e^{-1.5} = -1.75 \cdot 0.22313 \approx -0.390$.
Why not the others: (A) 0.112 replaces $\left(-\frac{1}{2}\right)^2 t^2$ by $\frac{1}{2}t^2$ in the second chain factor. (B) $0.335 = s'(3)$, the velocity. (C) -1.750 is the polynomial factor only; the factor $e^{-t/2}$ is lost.

20. Answer: C. $f'(x) = \dfrac{2x}{x^2+1} + \dfrac{1}{1+x^2} = \dfrac{2x+1}{x^2+1}$. Solve $\dfrac{2x+1}{x^2+1} = 0.8$ with the calculator: $x \approx 2.596$ (the only positive solution, since f' rises from 1 to a maximum near $x = 0.618$ and then decreases to 0).
Why not the others: (A) 0.500 solves $\dfrac{2x}{x^2+1} = 0.8$, dropping the arctangent term. (B) 1.225 solves $\dfrac{2}{1+x^2} = 0.8$, missing the chain factor $2x$ in the logarithm term. (D) 0.570 solves $f(x) = 0.8$, the *value* instead of the slope.

21. Answer: B. $\ln y = \sin x \ln x$, so $\dfrac{dy}{dx} = x^{\sin x}\left(\cos x \ln x + \dfrac{\sin x}{x}\right)$. At $x = 2$: $2^{\sin 2} \approx 1.8782$ and the bracket is $\cos 2 \ln 2 + \dfrac{\sin 2}{2} \approx -0.2884 + 0.4546 = 0.1662$. Product ≈ 0.312.
Why not the others: (A) 0.166 is the bracket alone: the factor y is forgotten. (C) $0.854 = \sin x \cdot x^{\sin x - 1}$, the power rule applied to a variable exponent. (D) $1.878 = y$ itself, with no derivative taken.

Part C: Free Response

Scoring guide style

Question 22 (9 points)

(a) *[2 points]*

$\tan\theta = \dfrac{h(t)}{300} = \dfrac{10t^2}{300} = \dfrac{t^2}{30}$, so $\theta(t) = \arctan\left(\dfrac{t^2}{30}\right)$. *1 point: expression for $\theta(t)$*

$\theta'(t) = \dfrac{1}{1+\frac{t^4}{900}} \cdot \dfrac{2t}{30} = \dfrac{60t}{900+t^4}$, so $\theta'(5) = \dfrac{300}{1525} \approx 0.197$ radians per second. *1 point: $\theta'(5)$ with units*

(b) *[2 points]*

T is the inverse of h. When $h = 1000$: $10t^2 = 1000$, $t = 10$, so $T(1000) = 10$. *1 point: uses the inverse relation, $T'(1000) = \dfrac{1}{h'(10)}$*

$h'(t) = 20t$, $h'(10) = 200$, so $T'(1000) = \dfrac{1}{200} = 0.005$ seconds per metre. *1 point: value with units*

(c) *[3 points]*

$\theta''(t) = \dfrac{60(900+t^4) - 60t \cdot 4t^3}{(900+t^4)^2} = \dfrac{60(900-3t^4)}{(900+t^4)^2}$. *1 point: θ''*

$\theta''(t) = 0 \Rightarrow t^4 = 300$, so $t = 300^{1/4} \approx 4.162$. *1 point: critical point*

$\theta'' > 0$ for $0 < t < 4.162$ and $\theta'' < 0$ for $4.162 < t < 20$, so θ' increases and then decreases: it has an absolute maximum at $t \approx 4.162$. (Check: $\theta'(4.162) \approx 0.208$, while $\theta'(0) = 0$ and $\theta'(20) \approx 0.007$.) *1 point: justification*

(d) *[2 points]*

$D = \sqrt{300^2 + h^2}$, so $D'(t) = \dfrac{h\,h'}{\sqrt{300^2+h^2}}$. *1 point: derivative*

At $t = 5$: $h = 250$, $h' = 100$, $D = \sqrt{152500} \approx 390.512$, so $D'(5) = \dfrac{25000}{390.512} \approx 64.018$ metres per second. *1 point: value with units*

Question 23 (9 points)

(a) *[2 points]*

$2x + 2y + 2x\dfrac{dy}{dx} + 6y\dfrac{dy}{dx} = 0$ *1 point: differentiates (product rule and chain rule)*

$(2x+6y)\dfrac{dy}{dx} = -(2x+2y)$, so $\dfrac{dy}{dx} = -\dfrac{x+y}{x+3y}$. *1 point: solves for $\dfrac{dy}{dx}$*

(b) *[2 points]*

Slope at $(1,1)$: $-\frac{2}{4} = -\frac{1}{2}$. *1 point*

$y - 1 = -\frac{1}{2}(x-1)$. *1 point*

(c) *[3 points]*

A horizontal tangent requires $\frac{dy}{dx} = 0$, so $x + y = 0$ and $y = -x$. *1 point*

Substituting into C: $x^2 - 2x^2 + 3x^2 = 2x^2 = 6$, so $x = \pm\sqrt{3}$. *1 point*

The points are $(\sqrt{3}, -\sqrt{3})$ and $(-\sqrt{3}, \sqrt{3})$; at both, $x + 3y = -2x \neq 0$, so the slope is 0. *1 point*

(d) *[2 points]*

Quotient rule on $\frac{dy}{dx} = -\frac{x+y}{x+3y}$:

$$\frac{d^2y}{dx^2} = -\frac{(1+\frac{dy}{dx})(x+3y) - (x+y)(1+3\frac{dy}{dx})}{(x+3y)^2}.$$

At $(1,1)$ with $\frac{dy}{dx} = -\frac{1}{2}$: numerator $= \frac{1}{2}\cdot 4 - 2\cdot\left(-\frac{1}{2}\right) = 3$, so $\frac{d^2y}{dx^2} = -\frac{3}{16}$. *1 point: value*

Since $\frac{d^2y}{dx^2} < 0$ at $(1,1)$, C is concave down there. *1 point: conclusion with reason*

(Simplification: the numerator equals $2y - 2x\frac{dy}{dx}$, and substituting gives $\frac{d^2y}{dx^2} = -\frac{12}{(x+3y)^3}$, which also gives $-\frac{3}{16}$ at $(1,1)$.)

AP Calculus AB – Unit 4 – Solutions

Unit 4 Solutions: Contextual Applications of Differentiation

Topic 4.1 check

1. The units of W' are million liters per day. *At $t = 10$ days the volume of water in the reservoir is decreasing at a rate of 1.8 million liters per day.*

2. The surrounding values are $t = 1$ and $t = 3$: $s'(2) \approx \dfrac{s(3) - s(1)}{3 - 1} = \dfrac{170 - 55}{2} = 57.5$. At $t = 2$ hours the car is traveling forward at about 57.5 miles per hour.

3. $P'(x) = -x + 40$, so $P'(20) = 20$: when 20 items are sold the profit is increasing at $\$20$ per item. $P(20) = -200 + 800 - 150 = 450$ and $P(21) = -220.5 + 840 - 150 = 469.5$, so $P(21) - P(20) = 19.5$, very close to $P'(20) = 20$.

Topic 4.2 check

1. $v(t) = 2t - 6$, $a(t) = 2$. $v(1) = -4$, $a(1) = 2$. Since $v(1) < 0$ the particle moves left; v and a have opposite signs, so the speed is decreasing at $t = 1$. $v(t) = 0$ at $t = 3$, and v changes from negative to positive there, so the particle changes direction at $t = 3$.

2. $v(t) = t^2(t - 3)$, $a(t) = 3t^2 - 6t = 3t(t - 2)$. For $t > 0$: $v < 0$ on $(0, 3)$, $v > 0$ on $(3, \infty)$; $a < 0$ on $(0, 2)$, $a > 0$ on $(2, \infty)$. Same signs on $(0, 2)$ (both $-$) and on $(3, \infty)$ (both $+$). The speed is increasing on $(0, 2)$ and $(3, \infty)$.

3. $v(t) = 2t - 4 = 0$ at $t = 2$ (direction change). $x(0) = 0$, $x(2) = -4$, $x(5) = 5$. Displacement $= x(5) - x(0) = 5$. Total distance $= |-4 - 0| + |5 - (-4)| = 4 + 9 = 13$.

Topic 4.3 check

1. $L'(15) = -4$: at $t = 15$ minutes the queue is getting shorter at a rate of 4 people per minute. $L''(15) = 3$: at $t = 15$ the rate of change of the queue length is increasing at 3 people per minute per minute. Because L' is negative and increasing (toward 0), the queue is getting shorter *more slowly*.

2. $H'(t) = -7e^{-0.1t} < 0$ and $H''(t) = 0.7e^{-0.1t} > 0$ for all t since $e^{-0.1t} > 0$. The coffee is always cooling ($H' < 0$), but the cooling rate shrinks in size (H' increases toward 0): it cools fast at first, then more and more slowly, approaching $20°$C.

3. $R'(x) = 60 - 0.2x$, $R'(100) = 40$: at a sales level of 100 items, revenue is increasing at $\$40$ per item. $R(101) = 6060 - 1020.1 = 5039.9$ and $R(100) = 6000 - 1000 = 5000$, so $R(101) - R(100) = 39.9 \approx R'(100)$. $R'(x) = 0$ when $x = 300$.

Topic 4.4 check

1. $\dfrac{dV}{dt} = 3s^2 \dfrac{ds}{dt}$. At $s = 4$: $3(16)(0.5) = 24$ cm^3/s.

2. Differentiate with respect to t: $2x\dfrac{dx}{dt} + 2y\dfrac{dy}{dt} = 0$, so $\dfrac{dy}{dt} = -\dfrac{x}{y}\dfrac{dx}{dt}$.

3. $\dfrac{dS}{dt} = 8\pi r \dfrac{dr}{dt} = 8\pi(3)(2) = 48\pi \approx 150.796$ cm^2/s.

Topic 4.5 check

1. $A = s^2$, $\dfrac{dA}{dt} = 2s\dfrac{ds}{dt} = 2(8)(3) = 48$ cm^2/s (increasing).

2. $V = \frac{4}{3}\pi r^3$, $\dfrac{dV}{dt} = 4\pi r^2 \dfrac{dr}{dt}$: $-12 = 4\pi(4)\dfrac{dr}{dt}$, so $\dfrac{dr}{dt} = -\dfrac{3}{4\pi} \approx -0.239$ cm/min. The radius is decreasing at about 0.239 cm per minute.

3. $x^2 + y^2 = 100$, so $x\,x' + y\,y' = 0$. When $y = 8$: $x = 6$, and $y' = -1$. Then $6x' + 8(-1) = 0$, $x' = \dfrac{4}{3}$ ft/s. The bottom is moving away from the wall at $\frac{4}{3}$ foot per second.

Topic 4.6 check

1. $f(0) = 1$, $f'(x) = 2e^{2x}$, $f'(0) = 2$, so $L(x) = 1 + 2x$. Since $e^{0.06} = f(0.03)$, $e^{0.06} \approx L(0.03) = 1.06$. $f''(x) = 4e^{2x} > 0$: concave up, tangent line below the graph, so 1.06 is an *underestimate* (actual $1.0618\ldots$).
2. $L(x) = 10 + 0.4(x - 5)$, $g(4.5) \approx 10 + 0.4(-0.5) = 9.8$. Since $g'' < 0$, g is concave down and lies below its tangent line, so 9.8 is an *overestimate*.
3. $f'(x) = \frac{1}{3}x^{-2/3}$, $f'(27) = \frac{1}{3}\cdot\frac{1}{9} = \frac{1}{27}$, $f(27) = 3$. $L(x) = 3 + \frac{1}{27}(x - 27)$ and $\sqrt[3]{28} \approx L(28) = 3 + \frac{1}{27} = \frac{82}{27} \approx 3.037$. ($f'' < 0$, so this is slightly too large; the true value is $3.0366\ldots$.)

Topic 4.7 check

1. Form $\frac{0}{0}$: $\lim\limits_{x\to 0} \dfrac{\sin x}{2x}$, again $\frac{0}{0}$: $\lim\limits_{x\to 0} \dfrac{\cos x}{2} = \dfrac{1}{2}$.

2. Form $\frac{0}{0}$: $\lim\limits_{x\to 2} \dfrac{3x^2}{2x} = \dfrac{12}{4} = 3$. Check: $\dfrac{(x-2)(x^2+2x+4)}{(x-2)(x+2)} \to \dfrac{12}{4} = 3$.

3. Form $\frac{\infty}{\infty}$: $\lim\limits_{x\to\infty} \dfrac{2x}{e^x}$, still $\frac{\infty}{\infty}$: $\lim\limits_{x\to\infty} \dfrac{2}{e^x} = 0$. The exponential grows faster than any power of x, so the quotient tends to 0.

Part A – Multiple choice, no calculator

1 **Answer: C.** $H'(10) = -2.5$ is an instantaneous rate with units °C per minute and a negative sign (decreasing).
Why not the others: (A) treats the rate as a total change over 10 minutes; (B) confuses the derivative with the value $H(10)$; (D) turns the instantaneous rate into an average rate over $[0, 10]$.

2 **Answer: A.** The values surrounding $t = 5$ are $t = 3$ and $t = 7$: $N'(5) \approx \dfrac{66 - 38}{7 - 3} = \dfrac{28}{4} = 7$.
Why not the others: (B) uses the far endpoints: $\frac{74-20}{10} = 5.4$, an average rate over the whole table; (C) divides by 10 instead of 4: $\frac{28}{10}$; (D) forgets to divide by the change in t.

3 **Answer: B.** $x'(t) = t^2 - 4t + 3 = (t - 1)(t - 3)$. Moving left means $v < 0$: $(t - 1)(t - 3) < 0$ for $1 < t < 3$.
Why not the others: (A), (C) and (D) are intervals where $v > 0$; (D) is exactly the union of the two rightward intervals, from mixing up left and right.

4 **Answer: D.** $v(2) = 4 - 12 = -8 < 0$, so the particle moves left. $a(t) = 2t - 6$, $a(2) = -2 < 0$. Same signs, so the speed is increasing.
Why not the others: (C) assumes that a negative a means slowing down without comparing with the sign of v; (A) and (B) read the direction wrongly (a negative v is leftward).

5 **Answer: B.** The graph gives v and the slope gives a. On $(0, 2)$: slope -2, v falls from 2 to -2, crossing 0 at $t = 1$. On $(1, 2)$: $v < 0$ and $a < 0$ – speed increasing.
Why not the others: (A) $v > 0$, $a < 0$: slowing; (C) $v = -2$ constant, $a = 0$: constant speed (a negative velocity alone does not mean speeding up); (D) $v < 0$, $a > 0$: slowing as v climbs toward 0.

6 **Answer: D.** $A' > 0$: A is increasing. $A'' < 0$: A' is decreasing, so the rate of increase is falling.
Why not the others: (A) reverses both signs; (B) reads $A' > 0$ as decreasing; (C) reads $A'' < 0$ as an increasing rate.

7 **Answer: A.** $\dfrac{dV}{dt} = 4\pi r^2 \dfrac{dr}{dt}$: $72\pi = 4\pi(9)\dfrac{dr}{dt} = 36\pi\dfrac{dr}{dt}$, so $\dfrac{dr}{dt} = 2$.
Why not the others: (B) inverts the ratio ($\frac{36\pi}{72\pi}$); (C) uses $4\pi r$ instead of $4\pi r^2$ ($\frac{72\pi}{12\pi}$); (D) omits the factor 4 ($\frac{72\pi}{9\pi}$).

8 **Answer: C.** $x^2 + y^2 = 100 \Rightarrow x\,x' + y\,y' = 0$. At $x = 6$: $y = 8$. $6(0.5) + 8y' = 0 \Rightarrow y' = -\dfrac{3}{8}$.
Why not the others: (A) loses the minus sign (the top is moving down); (B) uses $\frac{x}{y}$ inverted ($\frac{y}{x} \cdot 0.5$); (D) forgets to divide by y (-6×0.5).

9 **Answer: B.** $A = \ell w$, $\dfrac{dA}{dt} = \ell' w + \ell w' = 3(6) + 10(-2) = 18 - 20 = -2$.
Why not the others: (A) drops the sign; (C) uses $w' = +2$ $(18 + 20)$, ignoring that the width is decreasing; (D) is the area $\ell w = 60$ itself, not a rate.

10 **Answer: D.** $f'(x) = \dfrac{1}{2\sqrt{x}}$, so $f'(16) = \dfrac{1}{8}$ and $L(x) = 4 + \dfrac{1}{8}(x - 16)$. $L(17) = 4 + \dfrac{1}{8} = \dfrac{33}{8}$.
Why not the others: (A) uses slope $\frac{1}{\sqrt{x}} = \frac{1}{4}$; (B) uses slope $\frac{1}{x} = \frac{1}{16}$; (C) uses slope $\frac{1}{2x} = \frac{1}{32}$ – three wrong derivatives of $\sqrt{x}$.

11 **Answer: A.** $L(3.5) = 5 + (-2)(0.5) = 4$. Since $f'' < 0$, the graph is concave down and lies below its tangent line, so the tangent line value 4 is greater than $f(3.5)$.
Why not the others: (B) reverses the concavity rule; (C) and (D) use $+2 \cdot 0.5$ (sign error in $f'(3)$), giving 6.

12 **Answer: C.** Form $\frac{0}{0}$. L'Hôpital: $\lim\limits_{x \to 0} \dfrac{3\cos 3x}{5} = \dfrac{3}{5}$.
Why not the others: (A) treats the numerator tending to 0 as making the limit 0; (B) inverts the result; (D) multiplies $3 \cdot 5$.

13 **Answer: D.** Form $\frac{\infty}{\infty}$: $\lim \dfrac{e^x}{2x}$, again $\frac{\infty}{\infty}$: $\lim \dfrac{e^x}{2} = \infty$.
Why not the others: (A) assumes a polynomial beats the exponential; (B) cancels the x and keeps the coefficient $\frac{1}{2}$; (C) treats $\frac{\infty}{\infty}$ as 1.

14 **Answer: B.** Form $\frac{0}{0}$: $\lim\limits_{x \to 4} \dfrac{\frac{1}{2\sqrt{x}}}{1} = \dfrac{1}{2 \cdot 2} = \dfrac{1}{4}$.
Why not the others: (A) treats the numerator $\to 0$ as the limit; (C) uses $\frac{1}{\sqrt{x}}$ as the derivative of $\sqrt{x}$; (D) inverts the answer.

Part B – Multiple choice, calculator allowed

15 **Answer: C.** $P'(t) = 1200(0.04)e^{0.04t} = 48e^{0.04t}$; $P'(5) = 48e^{0.2} \approx 58.627$ bacteria per hour.
Why not the others: (A) 48 is $P'(0)$, the exponential was not evaluated; (B) $P(6) - P(5) = 59.816$ is an average change over one hour, not the instantaneous rate; (D) $P(5) = 1465.683$ is the population itself.

16 **Answer: A.** $v(2) = 2\cos 2 \approx -0.832$. $a(t) = \cos t - t \sin t$, so $a(2) \approx -0.416 - 1.819 = -2.235$. Same sign, so the speed is increasing.
Why not the others: (B) and (C) use the sign of only one of v, a; (D) compares magnitudes, which says nothing about speeding up or slowing down.

17 **Answer: D.** $C'(x) = 16x^{-0.2}$; $C'(50) = 16(50)^{-0.2} \approx 7.317$ dollars per pound.
Why not the others: (A) $50^{0.8}$ alone; (B) $16x^{0.8}$ – the exponent was not lowered by 1; (C) $C(50) = 957.305$ is the cost, not its rate.

18 **Answer: B.** $V' = 4\pi r^2 r'$: $25 = 4\pi(36)r' \Rightarrow r' = \dfrac{25}{144\pi} \approx 0.0553$. $S' = 8\pi r\, r' = 8\pi(6)(0.05526) \approx 8.333$ cm^2/s.
Why not the others: (A) is r' (the radius rate); (C) is the given V'; (D) $8\pi r = 150.796$ omits the factor r'.

19 **Answer: A.** $\tan\theta = \dfrac{h}{500}$, $\sec^2\theta\, \theta' = \dfrac{h'}{500}$. At $h = 1200$: $\sec^2\theta = 1 + 2.4^2 = 6.76$. $\theta' = \dfrac{80}{500(6.76)} \approx 0.024$ rad/s.
Why not the others: (B) multiplies by $\sin^2\theta$ instead of $\cos^2\theta$; (C) $\frac{80}{500} = 0.160$ ignores $\sec^2\theta$; (D) multiplies by $\sec^2\theta$ instead of dividing.

20 **Answer: C.** $f(1) = \ln 4 \approx 1.3863$, $f'(x) = \dfrac{2x}{x^2 + 3}$, $f'(1) = \dfrac{1}{2}$. $L(1.2) = 1.3863 + 0.5(0.2) = 1.4863 \approx 1.486$.

Why not the others: (A) is only $f(1)$; (B) is the true value $f(1.2) \approx 1.491$ rather than the linear approximation; (D) uses slope 1 instead of $f'(1) = \frac{1}{2}$.

21 **Answer: D.** Form $\frac{0}{0}$: $\lim_{x\to 0} \dfrac{\frac{3}{1+3x}}{2\cos 2x} = \dfrac{3}{2}$.
Why not the others: (A) treats the numerator $\to 0$ as the limit; (B) inverts the quotient; (C) differentiates the numerator only and forgets the 2 in the denominator's derivative.

Part C – Free response

22 **Scoring guide (9 points)**
$v(t) = x'(t) = 4e^{-t/2} - 2te^{-t/2} = e^{-t/2}(4-2t), \quad a(t) = v'(t) = e^{-t/2}(t-4).$
(a) 2 points.
1 point: $v(3) = e^{-1.5}(4-6) \approx -0.446$ and $a(3) = e^{-1.5}(3-4) \approx -0.223$.
1 point: The speed is *increasing* at $t = 3$ because $v(3) < 0$ and $a(3) < 0$ have the same sign.
(b) 2 points.
1 point: $v(t) = 0 \Rightarrow 4 - 2t = 0$ (since $e^{-t/2} > 0$), so $t = 2$ is the only candidate.
1 point: Justification: $v(t) > 0$ for $0 < t < 2$ and $v(t) < 0$ for $2 < t < 8$, so v changes from positive to negative at $t = 2$. The particle changes direction only at $t = 2$.
(c) 3 points.
1 point: Solve $4te^{-t/2} = 1$ on the calculator: $t \approx 0.289$ seconds.
1 point: $v(0.289) = e^{-0.1445}(4 - 0.578) \approx 2.962$.
1 point: Interpretation with units: *At $t = 0.289$ seconds the particle is moving in the positive direction (to the right) at 2.962 centimeters per second.*
(d) 2 points.
1 point: $x(7) = 28e^{-3.5} \approx 0.846 > 0$, so the particle is to the right of the origin.
1 point: $v(7) = e^{-3.5}(4-14) \approx -0.302 < 0$, so the particle is moving left, toward the origin. Because the position is positive and decreasing, the particle is moving *toward* the origin.

23 **Scoring guide (9 points)**
(a) 2 points.
1 point: By similar triangles, $\dfrac{r}{h} = \dfrac{6}{12}$, so $r = \dfrac{h}{2}$.
1 point: $V = \dfrac{1}{3}\pi\left(\dfrac{h}{2}\right)^2 h = \dfrac{1}{3}\pi \cdot \dfrac{h^2}{4} \cdot h = \dfrac{\pi}{12}h^3$.
(b) 3 points.
1 point: $\dfrac{dV}{dt} = \dfrac{\pi}{12} \cdot 3h^2 \dfrac{dh}{dt} = \dfrac{\pi h^2}{4}\dfrac{dh}{dt}$.
1 point: Substitute $h = 6$ and $\dfrac{dV}{dt} = -3\pi$: $-3\pi = \dfrac{\pi \cdot 36}{4}\dfrac{dh}{dt} = 9\pi\dfrac{dh}{dt}$.
1 point: $\dfrac{dh}{dt} = -\dfrac{1}{3}$ feet per minute (the depth is decreasing).
(c) 2 points.
1 point: The tangent line at $t = t_0$ has slope $h'(t_0) = -\frac{1}{3}$ and passes through $(t_0, 6)$: $y - 6 = -\dfrac{1}{3}(t - t_0)$.
1 point: $h(t_0 + 1) \approx 6 - \dfrac{1}{3} = \dfrac{17}{3} \approx 5.667$ ft.
(d) 2 points.
1 point: From part (b), $\dfrac{dh}{dt} = \dfrac{-3\pi}{\pi h^2/4} = -\dfrac{12}{h^2}$ for all $h > 0$. Differentiating, $\dfrac{d^2h}{dt^2} = 24h^{-3}\dfrac{dh}{dt} = 24h^{-3}\left(-\dfrac{12}{h^2}\right) = -\dfrac{288}{h^5} < 0$. So the graph of h is concave down.
1 point: A concave-down graph lies below its tangent lines, so the tangent line approximation $\dfrac{17}{3}$ is *greater than* the actual depth at $t_0 + 1$.

Unit 5 Solutions: Analytical Applications of Differentiation

Topic 5.1 check

1. f is continuous on $[1,9]$ and differentiable on $(1,9)$. Average rate: $\dfrac{3-1}{9-1}=\dfrac{1}{4}$. Solve $f'(c)=\dfrac{1}{2\sqrt{c}}=\dfrac{1}{4}$: $\sqrt{c}=2$, $c=4$, which lies in $(1,9)$. **Answer:** $c=4$.
2. g is differentiable, hence continuous on $[1,4]$ and differentiable on $(1,4)$. By the MVT there is c in $(1,4)$ with $g'(c)=\dfrac{g(4)-g(1)}{4-1}=\dfrac{-5-4}{3}=-3$.
3. h has a vertical asymptote at $x=2$, which is in $[0,4]$, so h is not continuous on $[0,4]$ and the hypothesis fails. The average rate is $\dfrac{h(4)-h(0)}{4}=\dfrac{\frac{1}{2}-(-\frac{1}{2})}{4}=\dfrac{1}{4}$, but $h'(x)=-\dfrac{1}{(x-2)^2}<0$ for every $x\neq 2$, so $h'(c)=\dfrac{1}{4}$ has no solution. The theorem is not contradicted because its hypothesis fails.

Topic 5.2 check

1. $f'(x)=4x^3-12x^2=4x^2(x-3)$, defined everywhere, zero at $x=0$ and $x=3$. **Answer:** $x=0$ **and** $x=3$.
2. $g'(x)=\dfrac{2}{3}(x-1)^{-1/3}$ is never 0 and is undefined at $x=1$, where $g(1)=0$ is defined. So $x=1$ is the only critical point. The graph has a cusp at $(1,0)$; since $g(x)\geq 0$ everywhere, it is the lowest point (a relative and absolute minimum), and the tangent line is vertical there.
3. False. The Extreme Value Theorem requires a *closed* interval. For example $f(x)=x$ is continuous on $(0,5)$ but has neither a maximum nor a minimum value there, since the endpoint values are not attained.

Topic 5.3 check

1. $f'(x)=3x^2-12=3(x-2)(x+2)$, positive for $x<-2$ and for $x>2$. f is increasing on $(-\infty,-2)$ and $(2,\infty)$ (and decreasing on $(-2,2)$).
2. $(x-1)^2\geq 0$, so the sign of f' is the sign of $x+3$ (except at $x=1$ where $f'=0$). $f'<0$ for $x<-3$ and $f'>0$ for $x>-3$, $x\neq 1$. So f is increasing on $[-3,\infty)$. The zero at $x=1$ does not matter: f' does not change sign there, so f keeps increasing through it.
3. $g'(x)=-2xe^{-x^2}$. Since $e^{-x^2}>0$, $g'>0$ when $x<0$ and $g'<0$ when $x>0$. g is increasing on $(-\infty,0)$ and decreasing on $(0,\infty)$.

Topic 5.4 check

1. $f'(x)=4x^3-16x=4x(x-2)(x+2)$. Test: $f'(-3)=-60<0$, $f'(-1)=12>0$, $f'(1)=-12<0$, $f'(3)=60>0$. f' changes from negative to positive at $x=-2$ and $x=2$, so f has relative minima there ($f(\pm 2)=-16$). f' changes from positive to negative at $x=0$: relative maximum, $f(0)=0$.
2. Critical points $x=-1$ and $x=2$. For $x<-1$: $(x+1)<0$, $(x-2)^2>0$, so $f'<0$; for $-1<x<2$ and $x>2$: $f'>0$. f' changes from negative to positive at $x=-1$: relative minimum. At $x=2$ f' is positive on both sides: no extremum.
3. $f'(x)=\dfrac{(x^2+1)-2x^2}{(x^2+1)^2}=\dfrac{(1-x)(1+x)}{(x^2+1)^2}$. The sign of f' is that of $(1-x)(1+x)$: negative for $x<-1$, positive on $(-1,1)$, negative for $x>1$. At $x=-1$, f' changes from negative to positive: relative minimum ($f(-1)=-\frac{1}{2}$). At $x=1$, f' changes from positive to negative: relative maximum ($f(1)=\frac{1}{2}$).

Topic 5.5 check

1. $f'(x)=3x^2-3=0$ at $x=\pm 1$, both in $(-2,3)$. Candidates: $f(-2)=-2$, $f(-1)=2$, $f(1)=-2$, $f(3)=18$. Absolute maximum value 18 (at $x=3$); absolute minimum value -2 (at $x=-2$ and $x=1$).
2. $g'(x)=e^{-x}(1-x)=0$ at $x=1$. Candidates: $g(0)=0$, $g(1)=\dfrac{1}{e}\approx 0.368$, $g(3)=3e^{-3}\approx 0.149$. Absolute maximum value $\dfrac{1}{e}$ (at $x=1$); absolute minimum value 0 (at $x=0$).
3. The Candidates Test is built on the Extreme Value Theorem, which needs a closed interval; $(0,2)$ is open. Here $f'(x)=2x>0$ on $(0,2)$ so there are no critical points, f is increasing, and the values approach 0 and 4 without

reaching them: f has no absolute maximum (and no absolute minimum) on $(0, 2)$.

Topic 5.6 check

1. $f'(x) = 3x^2 - 12x$, $f''(x) = 6x - 12$. $f'' < 0$ for $x < 2$ and $f'' > 0$ for $x > 2$. Concave down on $(-\infty, 2)$, concave up on $(2, \infty)$; since f'' changes sign at $x = 2$ there is an inflection point at $x = 2$ (the point $(2, -11)$).
2. $h''(x) = 30x^4 \geq 0$, which is 0 at $x = 0$ but positive on both sides. No sign change, so no inflection point.
3. The graph of f' is decreasing for $x < 3$ and increasing for $x > 3$. Hence f is concave down on $(-\infty, 3)$ and concave up on $(3, \infty)$, with an inflection point at $x = 3$.

Topic 5.7 check

1. $f'(x) = 3x^2 - 12 = 0$ at $x = \pm 2$; $f''(x) = 6x$. $f''(-2) = -12 < 0$: relative maximum at $x = -2$ ($f(-2) = 16$). $f''(2) = 12 > 0$: relative minimum at $x = 2$ ($f(2) = -16$).
2. $f'(x) = (x+1)e^x = 0$ at $x = -1$. $f''(x) = (x+2)e^x$, so $f''(-1) = e^{-1} > 0$. Since $f'(-1) = 0$ and $f''(-1) > 0$, f has a relative minimum at $x = -1$ (value $-\frac{1}{e}$).
3. $g'(x) = 4(x-2)^3$ and $g''(x) = 12(x-2)^2$, so $g'(2) = 0$ and $g''(2) = 0$: the test is inconclusive. By the First Derivative Test, $g' < 0$ for $x < 2$ and $g' > 0$ for $x > 2$, so g has a relative minimum at $x = 2$ (in fact an absolute minimum, $g(2) = 0$).

Topic 5.8 check

1. f rises for $x < 2$ and falls for $x > 2$, so $f' > 0$ for $x < 2$, $f'(2) = 0$, $f' < 0$ for $x > 2$. The parabola is concave down everywhere, so f' is decreasing: the graph of f' is a decreasing line crossing the x-axis at $x = 2$.
2. f decreases with slope -1 before $x = 3$ and increases with slope 2 after $x = 3$: a relative minimum at $x = 3$ with a corner. The one-sided slopes -1 and 2 differ, so f is *not* differentiable at $x = 3$ ($f'(3)$ does not exist).
3. f' has a relative minimum at $x = 4$, so f' changes from decreasing to increasing there. That means f'' changes from negative to positive, so f changes from concave down to concave up: f has a point of inflection at $x = 4$.

Topic 5.9 check

1. Table (a): changes $-3, -2, -1$. The values are decreasing ($f' < 0$) and the slopes are increasing from -3 to -1, so the graph is concave up ($f'' > 0$). Table (b): changes $-2, -3, -4$: decreasing and concave down. So the answer is **(a)**.
2. $f'(x) = (x-1)(x-5)$ is positive for $x < 1$ and $x > 5$: f is increasing on $(-\infty, 1)$ and $(5, \infty)$. $f''(x) = 2x - 6 > 0$ for $x > 3$: f is concave up on $(3, \infty)$.
3. $f'(x) = 3 > 0$, so f is increasing at a constant rate; $f''(x) = 0$ for all x. The graph of f is a straight line with slope 3 ($f(x) = 3x + C$): it has no concavity and no extrema or inflection points.

Topic 5.10 check

1. If the width is w, the length is ℓ with $2w + 2\ell = 40$, so $\ell = 20 - w$ and $A(w) = w(20 - w)$. Domain $0 \leq w \leq 20$ (the endpoints give a degenerate rectangle of area 0).
2. $x = 12 - y$, so $P(y) = (12 - y)y^2 = 12y^2 - y^3$ with $0 \leq y \leq 12$ (since $x \geq 0$ and $y \geq 0$).
3. A point on the parabola is (x, x^2). $D(x) = (x - 3)^2 + (x^2 - 0)^2 = (x - 3)^2 + x^4$, domain all real numbers.

Topic 5.11 check

1. $xy = 36$, so $S(x) = x + \dfrac{36}{x}$ for $x > 0$. $S'(x) = 1 - \dfrac{36}{x^2} = 0$ at $x = 6$. $S''(x) = \dfrac{72}{x^3} > 0$ on the domain, so S is concave up and $x = 6$ gives the absolute minimum. The numbers are 6 and 6, and the least sum is 12.
2. $V(x) = x(12-2x)^2$ on $[0, 6]$. $V'(x) = (12-2x)(12-6x) = 0$ at $x = 6$ and $x = 2$. $V(0) = 0$, $V(2) = 2 \cdot 8^2 = 128$, $V(6) = 0$. The maximum volume is 128 cm^3 (cut squares of side 2 cm).
3. Minimize $D(x) = x^2 + (x^2 - 3)^2 = x^4 - 5x^2 + 9$. $D'(x) = 4x^3 - 10x = 2x(2x^2 - 5) = 0$ at $x = 0$, $x = \pm\sqrt{5/2}$. D' is negative for $x < -\sqrt{5/2}$, positive on $(-\sqrt{5/2}, 0)$, negative on $(0, \sqrt{5/2})$, positive beyond: relative minima at $x = \pm\sqrt{5/2}$, where $D = \dfrac{25}{4} - \dfrac{25}{2} + 9 = \dfrac{11}{4}$, versus $D(0) = 9$ at the relative maximum; as $|x| \to \infty$, $D \to \infty$.

So the closest points are $\left(\pm\sqrt{5/2},\ \frac{5}{2}\right) \approx (\pm 1.581,\ 2.5)$, at distance $\sqrt{11/4} = \dfrac{\sqrt{11}}{2} \approx 1.658$.

Topic 5.12 check

1. $2x+2y\dfrac{dy}{dx}-4+2\dfrac{dy}{dx}=0$, so $\dfrac{dy}{dx}=\dfrac{2-x}{y+1}$. *Horizontal:* $x=2$: $4+y^2-8+2y=11 \Rightarrow y^2+2y-15=0 \Rightarrow y=3$ or $y=-5$: points $(2,3)$ and $(2,-5)$. *Vertical:* $y=-1$: $x^2+1-4x-2=11 \Rightarrow x^2-4x-12=0 \Rightarrow x=6$ or $x=-2$: points $(6,-1)$ and $(-2,-1)$.

2. $y+x\dfrac{dy}{dx}=0$, so $\dfrac{dy}{dx}=-\dfrac{y}{x}$. Then $\dfrac{d^2y}{dx^2}=-\dfrac{\frac{dy}{dx}\,x-y}{x^2}=-\dfrac{-y-y}{x^2}=\dfrac{2y}{x^2}$. At $(2,2)$: $\dfrac{2\cdot 2}{4}=1$. (Check: $y=\dfrac{4}{x}$ gives $y''=\dfrac{8}{x^3}=1$ at $x=2$.)

3. $2x-y-x\dfrac{dy}{dx}+2y\dfrac{dy}{dx}=0$, so $\dfrac{dy}{dx}=\dfrac{y-2x}{2y-x}$. Horizontal: $y=2x$; then $x^2-2x^2+4x^2=3x^2=3$, $x=\pm 1$: points $(1,2)$ and $(-1,-2)$.

Solutions: Part A Multiple choice, no calculator

Questions 1–14

1. Answer: A. *1.* $f'(x)=6x^2-18x+12=6(x-1)(x-2)$. *2.* $f'>0$ for $x<1$ and for $x>2$; $f'<0$ on $(1,2)$. *3.* f is increasing on $(-\infty,1)\cup(2,\infty)$.

Why not the others: (B) is the interval where f is *decreasing* (sign reversed). (C) and (D) each include part of $(1,2)$, where $f'<0$: the student used only one critical point.

2. Answer: B. *1.* Average rate: $\dfrac{f(2)-f(0)}{2}=\dfrac{6-0}{2}=3$. *2.* $f'(x)=3x^2-1=3 \Rightarrow x^2=\dfrac{4}{3} \Rightarrow c=\dfrac{2}{\sqrt{3}}$ (the positive root, in $(0,2)$).

Why not the others: (A) $\frac{4}{3}$ is c^2: forgot the square root. (C) 1 is the midpoint of the interval, not the solution of $f'(c)=3$ (indeed $f'(1)=2$). (D) $\sqrt{7/3}$ solves $f'(c)=6$: the student used $f(2)-f(0)$ without dividing by 2.

3. Answer: C. *1.* Critical points: $x=-3,1,2$. *2.* $(x-1)^2\geq 0$, so the sign of f' is that of $(x+3)(x-2)$: positive for $x<-3$, negative on $(-3,2)$ (with $f'(1)=0$ but still negative on both sides), positive for $x>2$. *3.* f' changes from negative to positive only at $x=2$: relative minimum there.

Why not the others: (A) At $x=-3$, f' changes from positive to negative: a relative *maximum*. (B) and (D) treat $x=1$ as a minimum, but f' does not change sign at a double root.

4. Answer: D. *1.* $f'(3)=0$ and $f''(3)=-4<0$. *2.* By the Second Derivative Test, f has a relative maximum at $x=3$.

Why not the others: (A) reverses the sign rule (that would need $f''(3)>0$). (B) An inflection point needs f'' to change sign; $f''(3)\neq 0$. (C) The test is inconclusive only when $f''(3)=0$.

5. Answer: C. *1.* $f''=0$ at $x=-1,0,2$. *2.* $(x-2)^2\geq 0$, so the sign of f'' is that of $x(x+1)$ for $x\neq 2$: positive for $x<-1$, negative on $(-1,0)$, positive for $x>0$ (also on both sides of 2). *3.* f'' changes sign at $x=-1$ and $x=0$ only.

Why not the others: (A) misses $x=-1$. (B) and (D) include $x=2$, a double root where f'' does not change sign (and $f''=0$ alone is not enough).

6. Answer: A. *1.* f is concave up where f' is increasing. *2.* The graph of f' rises from $(3,-2)$ to $(5,0)$, so f' is increasing on $(3,5)$.

Why not the others: (B) $(1,5)$ is where $f'<0$: f is decreasing, not concave up. (C) $(0,3)$ is where f' is decreasing, so f is concave *down*. (D) $(0,1)$ is where $f'>0$: f is increasing.

7. Answer: D. *1.* $f'(5)=0$, but $f'<0$ on both sides of $x=5$ (on $(3,5)$ it climbs to 0; on $(5,7)$ it falls from 0). *2.* No sign change of f', so no relative extremum. *3.* f' is increasing on $(3,5)$ and decreasing on $(5,7)$, so the concavity of f changes from up to down at $x=5$: a point of inflection.

Why not the others: (A), (B) Take $f'(5) = 0$ as proof of an extremum; there is no sign change. (C) Correctly sees no extremum but misses that f' changes from increasing to decreasing, so the concavity changes.

8. Answer: B. *1.* $f'(x) = 3x^2 - 6x = 3x(x-2) = 0$ at $x = 0, 2$. *2.* $f(-1) = -4$, $f(0) = 0$, $f(2) = -4$, $f(4) = 16$. *3.* The smallest value is -4.

Why not the others: (A) -2 is $f(1)$, an arbitrary interior value that is not a candidate. (C) 0 is the relative maximum value. (D) 16 is the absolute *maximum*.

9. Answer: C. *1.* $x = 12 - y$, so $P(y) = (12-y)y^2 = 12y^2 - y^3$ for $0 \le y \le 12$. *2.* $P'(y) = 24y - 3y^2 = 3y(8-y) = 0$ at $y = 0$ or $y = 8$. *3.* $P(0) = 0$, $P(8) = 4 \cdot 64 = 256$, $P(12) = 0$. Maximum 256.

Why not the others: (A) 216 assumes the symmetric split $x = y = 6$. (B) 128 comes from $y = 4$, $x = 8$: the roles of x and y were swapped. (D) 0 is the value at the endpoints.

10. Answer: A. *1.* Implicit differentiation: $2x + y + (x + 2y)\dfrac{dy}{dx} = 0$, so $\dfrac{dy}{dx} = -\dfrac{2x+y}{x+2y}$. *2.* A horizontal tangent needs $2x + y = 0$: at $(2, -4)$, $2 \cdot 2 - 4 = 0$ and the denominator is $2 - 8 = -6 \neq 0$.

Why not the others: (B) At $(4, -2)$ the denominator $x + 2y = 0$: the tangent is *vertical*. (C) At $(2, 2)$ the slope is -1. (D) At $(0, 2\sqrt{3})$ the slope is $-\dfrac{1}{2}$.

11. Answer: D. *1.* f is differentiable, so the MVT applies on $[1, 5]$. *2.* $\dfrac{f(5) - f(1)}{5 - 1} = \dfrac{12}{4} = 3$, so $f'(c) = 3$ for some c in $(1, 5)$.

Why not the others: (A) assumes the tangent at the midpoint has the average slope. (B) assumes a constant slope; only the average is 3. (C) 12 is the total change $f(5) - f(1)$, not divided by 4.

12. Answer: B. *1.* $f' > 0$ means the values increase; $f'' < 0$ means the increments get smaller. *2.* Table (B): increments $4, 3, 2$: increasing, with decreasing increments.

Why not the others: (A) increments $3, 4, 5$: increasing but concave up. (C) values decrease ($f' < 0$). (D) values decrease with increments $-6, -3, -1$: decreasing and concave up.

13. Answer: A. *1.* $f''(x) = 6ax + 2b$. An inflection point at $x = 1$ needs $f''(1) = 6a + 2b = 0$, so $b = -3a$. *2.* The point is on the graph: $f(1) = a + b = 4$. *3.* $a - 3a = 4$, so $a = -2$ and $b = 6$.

Why not the others: (B) -2 is the value of a, not b. (C) 4 is $f(1) = a + b$ mistaken for b. (D) 12 comes from using $f'(1) = 0$ ($3a + 2b = 0$) instead of $f''(1) = 0$, which gives $a = -8$, $b = 12$.

14. Answer: C. *1.* This is the First Derivative Test: f' changes from positive to negative at a point where f is continuous.

Why not the others: (A) $f'(c) = 0$ gives only a critical point (it could be a minimum or neither). (B) $f'(c) = 0$ and $f''(c) > 0$ guarantees a relative *minimum*. (D) f'' changing sign at c describes a point of inflection, not an extremum.

Solutions: Part B Multiple choice, graphing calculator allowed

Questions 15–21

15. Answer: B. *1.* $f'(x) = 4x^3 - 6x + 1$; a calculator gives the zeros $x \approx -1.301, 0.170, 1.131$. *2.* f' is negative for $x < -1.301$, positive on $(-1.301, 0.170)$, negative on $(0.170, 1.131)$, positive beyond. *3.* f' changes from positive to negative only at $x \approx 0.170$: the relative maximum.

Why not the others: (A) and (C) are the two relative *minima*. (D) 0.084 is the maximum *value* $f(0.170)$, not the x-coordinate.

16. Answer: D. *1.* f is continuous and differentiable, so the MVT applies. Average rate: $\dfrac{e^2 - 1}{4} \approx 1.597$. *2.* $f'(x) = \frac{1}{2}e^{x/2} = 1.597$ gives $e^{x/2} = 3.195$, so $c = 2\ln(3.195) \approx 2.323$.

Why not the others: (A) 1.597 is the average rate itself, not the x-value. (B) 2.000 is the midpoint. (C) 3.195 is $e^{c/2}$: the student forgot to take the logarithm.

17. Answer: C. *1.* g is increasing where $g' > 0$: $\ln(x^2+1) > 1$. *2.* $x^2+1 > e$, so $x^2 > e-1 \approx 1.718$, hence $|x| > 1.311$. *3.* $x < -1.311$ or $x > 1.311$.

Why not the others: (A) misses the negative solutions. (B) reverses the inequality (that is where g is decreasing). (D) $1.718 = e-1$ is x^2; the square root was not taken.

18. Answer: A. *1.* $f'(x) = \sin x + x\cos x = 0$ has the solution $x \approx 2.029$ in $(0,\pi)$. *2.* $f(0) = 0$, $f(2.029) \approx 1.820$, $f(\pi) = 0$. *3.* The maximum value is 1.820.

Why not the others: (B) 2.029 is where the maximum occurs, not its value. (C) 1.571 is $f(\pi/2)$, an arbitrary interior value. (D) 0 is the value at the endpoints (the minimum).

19. Answer: B. *1.* Minimize the squared distance $D(x) = x^2 + e^{2x}$. *2.* $D'(x) = 2x + 2e^{2x} = 0$; a calculator solves $x + e^{2x} = 0$: $x \approx -0.426$. *3.* $D' < 0$ to the left and $D' > 0$ to the right, so this is the minimum.

Why not the others: (A) $0.653 = e^{-0.426}$ is the y-coordinate. (C) 0.780 is the minimum distance itself. (D) $x = 0$ gives the point $(0,1)$, where $D'(0) = 2 \neq 0$.

20. Answer: D. *1.* $f'(x) = e^x - 3x^2 = 0$ at $x \approx -0.459,\ 0.910,\ 3.733$, all in $(-1,4)$. *2.* Candidates: $f(-1) = 1.368$, $f(-0.459) = 0.729$, $f(0.910) = 1.731$, $f(3.733) = -10.216$, $f(4) = -9.402$. *3.* The smallest is -10.216.

Why not the others: (A) -9.402 is the endpoint value $f(4)$: the student ignored the critical point near 3.733. (B) 0.729 is a relative minimum value but not the smallest. (C) 1.731 is the relative maximum value.

21. Answer: A. *1.* T is differentiable, so the MVT applies on $[0,12]$: $\dfrac{T(12)-T(0)}{12} = \dfrac{60}{12} = 5$. *2.* Some c in $(0,12)$ has $T'(c) = 5$.

Why not the others: (B) 5.25 is the average rate on $[3,7]$; the MVT gives a c somewhere in $(3,7)$, not specifically $t = 5$. (C) The data are increasing but nothing forces $T' > 0$ at every instant. (D) 60 is the total change $T(12) - T(0)$, not divided by 12.

Solutions: Part C Free response

Question 22 (calculator), Question 23 (no calculator)

Question 22 (9 points)

$C(t) = 6te^{-t/3}$, so $C'(t) = 6e^{-t/3}\left(1 - \dfrac{t}{3}\right)$.

(a) $C'(5) = 6e^{-5/3}\left(1 - \dfrac{5}{3}\right) = -4e^{-5/3} \approx -0.756$. *1 point:* $C'(5) = -0.756$ *(accept* -0.755 *or* -0.756*)*

At $t = 5$ hours the concentration of medicine is *decreasing* at a rate of 0.756 milligrams per litre per hour, because $C'(5) < 0$. *1 point: units and "decreasing" with the reason*

(b) $C'(t) = 0$ gives $1 - \dfrac{t}{3} = 0$, so $t = 3$. *1 point: critical point*

$C(0) = 0$, $C(3) = \dfrac{18}{e} \approx 6.622$, $C(10) = 60e^{-10/3} \approx 2.140$. *1 point: values at the critical point and endpoints*

C is continuous on $[0,10]$, so by the Candidates Test the greatest concentration occurs at $t = 3$ hours (about 6.622 mg/L). *1 point: answer with justification*

(c) $C''(t) = \dfrac{2}{3}e^{-t/3}(t-6)$. *1 point*

Since $e^{-t/3} > 0$, $C''(t) < 0$ exactly when $t < 6$. The graph of C is concave down on $(0,6)$ because $C''(t) < 0$ there. *1 point: interval and reason*

(d) The rate of change of concentration is $C'(t)$; it is decreasing most rapidly where C' is least. *1 point: recognizes the need to minimize* C'

$C''(t) < 0$ on $(0,6)$ and $C''(t) > 0$ on $(6,10)$, so C' decreases and then increases, with its minimum at $t = 6$. Check with the Candidates Test on C': $C'(0) = 6$, $C'(6) = -6e^{-2} \approx -0.812$, $C'(10) \approx -0.499$. The concentration is decreasing most rapidly at $t = 6$ hours. *1 point:* $t = 6$ *with justification*

Question 23 (9 points)

(a) $f'(-2) = 0$ and $f'(2) = 0$ are the only zeros (and f' exists everywhere on $(-4,6)$). *1 point:* $x = -2$ *maximum and* $x = 2$ *minimum*

f' changes from positive to negative at $x = -2$, so f has a relative maximum at $x = -2$. f' changes from negative to positive at $x = 2$, so f has a relative minimum at $x = 2$. *1 point: justification (sign change of f')*

(b) The graph of f' has corners at $x = 0$ and $x = 4$. f' is decreasing on $(-4, 0)$, increasing on $(0, 4)$, and decreasing on $(4, 6)$. *1 point: $x = 0$ and $x = 4$*

f' changes from decreasing to increasing at $x = 0$ and from increasing to decreasing at $x = 4$, so the concavity of f changes at both: points of inflection at $x = 0$ and $x = 4$. *1 point: reason*

(c) The candidates are the critical points $x = -2$ and $x = 2$ and the endpoints $x = -4$ and $x = 6$: *1 point: considers critical points and endpoints*

x	-4	-2	2	6
$f(x)$	1	3	-1	4

f is continuous on $[-4, 6]$, so the absolute minimum value is -1 (at $x = 2$) *1 point*
and the absolute maximum value is 4 (at $x = 6$). *1 point*

(d) $f(-4) = 1$ and $f'(-4) = 2$, so the tangent line is $y = 1 + 2(x + 4)$. At $x = -3$: $y = 3$, so $f(-3) \approx 3$. *1 point: equation and approximation*

f' is decreasing on $(-4, 0)$, so f is concave down there and the tangent line lies above the graph. The approximation 3 is *greater than* $f(-3)$ (in fact $f(-3) = 2.5$). *1 point: conclusion with reason*

Unit 6 Solutions

Topic 6.1 check

1. The graph of $r(t) = 2t$ on $[0, 5]$ is a triangle with base 5 and height $r(5) = 10$: area $\frac{1}{2}(5)(10) = 25$ litres.
2. $v(t) = 3 - t$ is zero at $t = 3$ with $v(0) = 3$ and $v(5) = -2$. Above the axis: $\frac{1}{2}(3)(3) = 4.5$. Below: $\frac{1}{2}(2)(2) = 2$. Displacement $= 4.5 - 2 = 2.5$ m; total distance $= 4.5 + 2 = 6.5$ m.
3. The number of loaves sold between 2 and 6 hours after the bakery opens.

Topic 6.2 check

1. $\Delta x = 1$. $L_4 = f(0) + f(1) + f(2) + f(3) = 0 + 1 + 1.414 + 1.732 = 4.146$. $R_4 = 1 + 1.414 + 1.732 + 2 = 6.146$. Since $\sqrt{x}$ is increasing, L_4 is an under-estimate (and R_4 an over-estimate); the exact area is $\frac{16}{3} \approx 5.333$.
2. Widths $2, 1, 4$: $\frac{2+5}{2}(2) + \frac{5+6}{2}(1) + \frac{6+3}{2}(4) = 7 + 5.5 + 18 = 30.5$.
3. L_n is an over-estimate and R_n an under-estimate: on each strip the left height is the larger one.

Topic 6.3 check

1. $\Delta x = \frac{1}{n}$ and $x_i = \frac{i}{n}$, so $a = 0$, $b = 1$: $\displaystyle\int_0^1 \sqrt{x}\,dx = \left[\tfrac{2}{3}x^{3/2}\right]_0^1 = \tfrac{2}{3}$.
2. The graph is the upper half of the circle of radius 2: $\frac{1}{2}\pi(2)^2 = 2\pi$.
3. $\Delta x = \frac{1}{2}$, $x_i = \frac{i}{2}$: $R_4 = \displaystyle\sum_{i=1}^{4} e^{i/2} \cdot \tfrac{1}{2} = \tfrac{1}{2}\left(e^{0.5} + e + e^{1.5} + e^2\right) \approx 8.119$.

Topic 6.4 check

1. $F'(x) = \ln(x^2 + 1)$, so $F'(2) = \ln 5 \approx 1.609$.
2. By the chain-rule form of the Fundamental Theorem, $\dfrac{d}{dx}\displaystyle\int_0^{x^3} \sin(t^2)\,dt = \sin\left((x^3)^2\right) \cdot 3x^2 = 3x^2 \sin(x^6)$.
3. $g'(6) = f(6) = -2$. $g''(3) = f'(3) = -1.5$ (slope of the segment from $(2, 3)$ to $(4, 0)$).

Topic 6.5 check

1. $g'(x) = x^2 - 4 = 0$ gives $x = 2$ for $x > 0$. g' changes from negative to positive at $x = 2$, so g has a relative minimum there. $g''(x) = 2x > 0$ for $x > 0$, so g is concave up on $(0, \infty)$.
2. $g' = f$ changes from $+$ to $-$ at $x = 3$ (relative maximum) and from $-$ to $+$ at $x = 7$ (relative minimum).
3. At $x = 1$ the semicircle is rising, so f is increasing and $g''(1) = f'(1) > 0$: g is concave up.

Topic 6.6 check

1. $\int_3^5 f = \int_1^5 f - \int_1^3 f = 8 - 3 = 5$. $\int_5^1 3f(x)\,dx = -3\int_1^5 f = -24$.
2. x^3 is odd, so $\int_{-3}^{3} x^3 dx = 0$; the second term is the area of a semicircle of radius 3: $\frac{9\pi}{2}$. Total $\frac{9\pi}{2}$.
3. On $[0, 1]$, $x^2 \geq x^3$, so $\int_0^1 x^2\,dx \geq \int_0^1 x^3\,dx$: yes ($\frac{1}{3} \geq \frac{1}{4}$).

Topic 6.7 check

1. $\left[\frac{x^3}{3} - x^2\right]_{-1}^{2} = \left(\frac{8}{3} - 4\right) - \left(-\frac{1}{3} - 1\right) = -\frac{4}{3} + \frac{4}{3} = 0$.
2. $\left[\tan x\right]_0^{\pi/3} = \sqrt{3} - 0 = \sqrt{3}$.
3. $V(3) = 10 + \int_1^3 3t^2 dt = 10 + \left[t^3\right]_1^3 = 10 + 26 = 36$ litres.

Topic 6.8 check

1. $x^4 + \dfrac{2}{x} + 3\sin x + C$ (since $-2x^{-2}$ integrates to $2x^{-1}$).
2. $\dfrac{(x+1)^2}{x} = x + 2 + \dfrac{1}{x}$, so $\dfrac{x^2}{2} + 2x + \ln|x| + C$.
3. $f'(x) = 3x^2 + C_1$, $f'(0) = 2 \Rightarrow C_1 = 2$. $f(x) = x^3 + 2x + C_2$, $f(0) = -1 \Rightarrow C_2 = -1$. So $f(x) = x^3 + 2x - 1$.

Topic 6.9 check

1. $u = x^3$, $du = 3x^2 dx$: $\frac{1}{3}\int \cos u\, du = \frac{1}{3}\sin(x^3) + C$.
2. $u = \ln x$, $du = \frac{dx}{x}$; $x = 1 \Rightarrow u = 0$, $x = e \Rightarrow u = 1$: $\int_0^1 u^2 du = \frac{1}{3}$.
3. $u = x^2 + 4$, $du = 2x\,dx$; limits 4 and 8: $\int_4^8 \frac{du}{u} = \ln 8 - \ln 4 = \ln 2$.

Topic 6.10 check

1. $\dfrac{x+5}{x-2} = 1 + \dfrac{7}{x-2}$, so $x + 7\ln|x-2| + C$.
2. $x^2 - 6x + 10 = (x-3)^2 + 1$, so $\arctan(x-3) + C$.

Topic 6.14 check

1. (a) $u = 1 + e^x$ gives $\ln(1+e^x) + C$. (b) Expand: $x^4 + 4x^2 + 4$, so $\frac{x^5}{5} + \frac{4x^3}{3} + 4x + C$. (c) Complete the square: $(x+2)^2 + 4$, so $\frac{1}{2}\arctan\frac{x+2}{2} + C$.
2. $\int e^{x^2} dx$ has no elementary antiderivative. ($\int xe^{x^2} dx = \frac{1}{2}e^{x^2} + C$ by substitution.)

Part A solutions — No calculator

Item 1. Answer: (B). $\left[x^2 + x\right]_1^3 = 12 - 2 = 10$.
Why not the others: (A) $8 = \int_1^3 2x\,dx$ drops the $+1$. (C) $12 = F(3)$ alone, forgetting to subtract $F(1)$. (D) $7 = f(3)$ evaluates the integrand instead of the antiderivative.

Item 2. Answer: (D). By the chain-rule form of the FTC, $G'(x) = \sqrt{x^2+5} \cdot 2x$. At $x = 2$ the inner function is $t = x^2 = 4$: $\sqrt{4+5} = 3$, so $G'(2) = 3 \cdot (2 \cdot 2) = 12$.
Why not: (A) 3 omits the factor $\frac{d}{dx}x^2 = 2x$. (B) 6 uses a chain factor of 2 instead of $2x = 4$. (C) $4\sqrt{7}$ evaluates the integrand at $t = x = 2$ instead of $t = x^2 = 4$.

Item 3. Answer: (A). Widths $2, 3, 1$: $\frac{4+7}{2}(2) + \frac{7+9}{2}(3) + \frac{9+11}{2}(1) = 11 + 24 + 10 = 45$.
Why not: (B) $38 = 4(2) + 7(3) + 9(1)$ is the left sum. (C) $52 = 7(2) + 9(3) + 11(1)$ is the right sum. (D) 90 omits the factor $\frac{1}{2}$ in each trapezoid.

Item 4. Answer: (C). $f(x) = 2x^3 - 2x^2 + C$; $f(1) = 0 + C = 2$, so $C = 2$ and $f(2) = 16 - 8 + 2 = 10$.
Why not: (A) 8 drops C. (B) 4 integrates $-4x$ as $-4x^2$ (no division by 2): then $C = 4$ and $f(2) = 16 - 16 + 4 = 4$. (D) $16 = f'(2)$ evaluates the derivative.

Item 5. Answer: (D). $u = 1 + 3x^2$, $x\,dx = \frac{1}{6}du$, limits $1 \to 4$: $\frac{1}{6} \cdot \frac{2}{3}\left[u^{3/2}\right]_1^4 = \frac{1}{9}(8 - 1) = \frac{7}{9}$.
Why not: (A) $\frac{14}{3}$ omits the factor $\frac{1}{6}$. (B) $\frac{14}{9}$ uses $\frac{1}{3}$ instead of $\frac{1}{6}$. (C) $\frac{1}{9}$ keeps the x-limits 0 and 1 on the u-integral.

Item 6. Answer: (A). Even: $\int_{-3}^3 f = 2\int_0^3 f = 8$. Constant: $\int_{-3}^3 2\,dx = 2(6) = 12$. Sum $= 20$.
Why not: (B) $16 = 4 + 12$ does not double $\int f$. (C) 8 omits the constant term. (D) $32 = 2(4 + 12)$ doubles the constant's integral a second time.

Item 7. Answer: (C). $g' = f$ and $g'' = f'$. g increasing needs $f > 0$: $[0, 2)$ and $(7, 8]$. g concave up needs $f' > 0$ (f rising): $(6, 8)$. Both hold on $(7, 8)$.
Why not: (A) $(0, 2)$: $f > 0$ but f is falling, so concave down. (B) $(2, 4)$: $f < 0$ and falling, so decreasing and concave down. (D) $(6, 7)$: f is rising but negative, so g is decreasing.

Item 8. Answer: (B). $v = 3(t-1)(t-3)$: positive on $[0,1)$, negative on $(1,3)$, positive on $(3,4]$. With $s(t) = t^3 - 6t^2 + 9t$: $s(0) = 0$, $s(1) = 4$, $s(3) = 0$, $s(4) = 4$. Distance $= |4-0| + |0-4| + |4-0| = 12$.
Why not: (A) 4 is the displacement $s(4) - s(0)$. (C) 8 omits one of the three legs. (D) 0 is the position (or net change) at $t = 3$.

Item 9. Answer: (D). $\Delta x = \frac{3}{n}$ and $x_k = 1 + k \cdot \frac{3}{n}$: $a = 1$, $b = 4$, $f(x) = x^2$. $\int_1^4 x^2 dx = \frac{64-1}{3} = 21$.
Why not: (A) $9 = \int_0^3 x^2 dx$ takes the limits from Δx alone (0 to 3). (B) 63 keeps an extra factor 3. (C) $\frac{64}{3} = \int_0^4 x^2 dx$ forgets that the sample points start at 1.

Item 10. Answer: (A). $u = x^2$, $x\,dx = \frac{1}{2}du$: $\frac{1}{2}\int e^u du = \frac{1}{2}e^{x^2} + C$.
Why not: (B) omits the $\frac{1}{2}$. (C) integrates the factors x and e^{x^2} separately. (D) divides by the derivative $2x$ as if for e^{ax}, which is valid only for a constant inner derivative.

Item 11. Answer: (C). Decreasing: $L_4 > I > R_4$. Concave up: the chords lie above the graph, so $T_4 > I$. And $T_4 = \frac{1}{2}(L_4 + R_4) < L_4$. Hence $R_4 < I < T_4 < L_4$.
Why not: (A) puts T_4 below I (true only for concave down). (B) is the order for an *increasing*, concave-up function. (D) puts I below R_4, but a right sum under-estimates a decreasing function.

Item 12. Answer: (B). $g'(x) = x^2 - 4x = x(x-4)$: positive for $x < 0$, negative on $(0,4)$, positive for $x > 4$. So g has a relative maximum at 0 and a relative minimum at 4.
Why not: (A) 0 is a relative maximum. (C) 2 is where $g''(x) = 2x - 4 = 0$, an inflection point. (D) 6 is a zero of g ($g(x) = \frac{x^3}{3} - 2x^2$), not of g'.

Item 13. Answer: (A). $u = 1 + e^x$, $du = e^x dx$, limits $2 \to 4$: $\int_2^4 \frac{du}{u} = \ln 4 - \ln 2 = \ln 2$.
Why not: (B) $\ln 4$ forgets to subtract the value at the lower limit. (C) $\frac{1}{4} = \frac{3}{4} - \frac{1}{2}$ evaluates the integrand at the limits. (D) $\ln 3$ drops the "1+" (treating the limits as $e^0 = 1$ and 3).

Item 14. Answer: (D). $u = 2x + 1$, $dx = \frac{1}{2}du$, limits $1 \to 5$: $\frac{1}{2}\int_1^5 f(u)\,du = \frac{1}{2}(12) = 6$.
Why not: (A) 12 omits $\frac{1}{2}$. (B) 24 multiplies by 2. (C) 4 divides by 3 (the sum of the coefficients) instead of by the derivative 2.

Part B solutions — Calculator allowed

Item 15. Answer: (C). The calculator gives $\int_0^2 e^{-x^2} dx \approx 0.882$.
Why not: (A) $0.018 = e^{-4}$ evaluates the integrand at 2. (B) $0.865 = \int_0^2 e^{-x} dx$ replaces x^2 by x. (D) 1.764 doubles the value (a spurious symmetry argument).

Item 16. Answer: (A). $3\frac{10.2+14.6}{2} + 2\frac{14.6+13.1}{2} + 3\frac{13.1+9.4}{2} + 4\frac{9.4+6.3}{2} = 37.2 + 27.7 + 33.75 + 31.4 = 130.05$.
Why not: (B) 136.7 is the left sum. (C) 123.4 is the right sum. (D) 260.1 omits the $\frac{1}{2}$.

Item 17. Answer: (D). $V'(t) = 0 \iff e^{-0.2t} = 0.4 \iff t = 5\ln 2.5 \approx 4.581$. $V' > 0$ for $t < 4.581$ and $V' < 0$ after, so V has its maximum there.
Why not: (A) $0.916 = \ln 2.5$ forgets to divide by 0.2. (B) 2.291 divides by 0.4. (C) 3.000 uses the linear approximation $e^{-0.2t} \approx 1 - 0.2t$.

Item 18. Answer: (B). $\Delta x = 1$, midpoints $0.5, 1.5, 2.5$: $\sqrt{1.125} + \sqrt{4.375} + \sqrt{16.625} \approx 1.0607 + 2.0917 + 4.0774 = 7.230$.
Why not: (A) 5.414 is the left sum (using $0, 1, 2$). (C) 9.706 is the right sum (using $1, 2, 3$). (D) 7.560 is the trapezoidal sum.

Item 19. Answer: (C). $f(2) = f(0) + \int_0^2 e^{\sin x} dx = 5 + 4.237 = 9.237$.
Why not: (A) 4.237 omits $f(0)$. (B) $2.483 = f'(2)$. (D) $7.483 = f(0) + f'(2)$.

Item 20. Answer: (A). $H'(x) = e^{-(x^2)^2} \cdot 2x = 2xe^{-x^4}$; $H'(1.2) = 2.4e^{-2.0736} \approx 0.302$.
Why not: (B) $0.126 = e^{-x^4}$ omits the chain factor. (C) $0.569 = 2xe^{-x^2}$ evaluates the integrand at $t = x$. (D)

$0.251 = 2e^{-x^4}$ uses 2 instead of $2x$.

Item 21. Answer: (D). $V(8) = 40 + \int_0^8 \big(8 + 3\sin(0.5t)\big)dt = 40 + 64 + 6(1 - \cos 4) = 113.92$ litres (calculator in radian mode).
Why not: (A) 73.92 omits the initial 40. (B) $45.73 = 40 + R(8)$ treats the rate as the amount. (C) 104.01 is the result with the calculator in degree mode ($\cos 4^\circ \approx 0.9976$).

Part C solutions

Scoring guide style

Question 22 (calculator active, 9 points).
(a) *1 point:* trapezoidal sum $\frac{3.0+5.0}{2}(2) + \frac{5.0+6.4}{2}(3) + \frac{6.4+5.6}{2}(2) + \frac{5.6+4.8}{2}(1) = 8 + 17.1 + 12 + 5.2 = 42.3$.
1 point: $\int_0^8 L(t)\,dt$ is the total amount of water, in thousands of litres, that flows out of the tank during the 8 hours $0 \le t \le 8$.
(b) *1 point:* $\int_0^8 \big(4 + 3\sin\frac{t}{2}\big)dt = 41.922$ (thousand litres). *1 point:* the amount at $t = 8$ is $25 + \int_0^8 E(t)\,dt - \int_0^8 L(t)\,dt$. *1 point:* $25 + 41.922 - 42.3 = 24.622$ thousand litres.
(c) *1 point:* The rate of change of the amount W is $W'(5) = E(5) - L(5) = 4 + 3\sin 2.5 - 6.4 = -0.605$. *1 point:* Since $W'(5) < 0$, the amount of water is decreasing at $t = 5$.
(d) *1 point:* $W'(2) = E(2) - L(2) = 4 + 3\sin 1 - 5 = 1.524 > 0$ and $W'(5) = -0.605 < 0$. *1 point:* E and L are continuous, so W' is continuous; by the Intermediate Value Theorem there is a t in $(2, 5)$ with $W'(t) = 0$. Yes, at that time the amount is neither increasing nor decreasing.

Question 23 (no calculator, 9 points). Areas: from -4 to -2 the triangle has area -2; the quarter disc on $[-2, 0]$ and on $[0, 2]$ each has area $\frac{1}{4}\pi(2)^2 = \pi$; from 2 to 4 the triangle has area -2.
(a) *1 point:* $g(-2) = \int_0^{-2} f = -\int_{-2}^0 f = -\pi$. *1 point:* $g(4) = \int_0^2 f + \int_2^4 f = \pi - 2$.
(b) *1 point:* $g'(4) = f(4) = -2$. *1 point:* $y - (\pi - 2) = -2(x - 4)$.
(c) *1 point:* $x = 0$ and $x = 4$. *1 point:* $g''(x) = f'(x)$. f increases for $x < 0$ and decreases on $(0, 2)$ (and on $(2, 4)$), so f' changes from positive to negative at $x = 0$; f decreases on $(2, 4)$ and increases on $(4, 6)$ (slope 2), so f' changes from negative to positive at $x = 4$. At $x = -2$ and $x = 2$, f' keeps its sign (positive, then negative, respectively), so there is no inflection.
(d) *1 point:* $g'(x) = f(x) = 0$ at $x = -2,\ 2,\ 5$. *1 point:* candidate values: $g(-4) = -\int_{-4}^0 f = 2 - \pi$; $g(-2) = -\pi$; $g(2) = \pi$; $g(5) = g(4) + \int_4^5 f = \pi - 2 - 1 = \pi - 3$; $g(6) = \pi - 2$. *1 point:* the least of these is $g(-2) = -\pi \approx -3.142$, so the absolute minimum value is $-\pi$ (at $x = -2$, where $g' = f$ changes from negative to positive).

Unit 7 Solutions: Differential Equations

Topic "Check yourself" answers

Topics 7.1–7.8

Topic 7.1 check.

1. $\dfrac{dN}{dt} = k\,(5000 - N)$ (equivalently $k(N - 5000)$ with the opposite sign of k).
2. Sugar in: $4 \cdot 0.3 = 1.2$ kg/min. Sugar out: $4 \cdot \dfrac{S}{200} = \dfrac{S}{50}$ kg/min. So $\dfrac{dS}{dt} = 1.2 - \dfrac{S}{50}$.
3. $\dfrac{dP}{dt} = 0 \Rightarrow 0.04P = 20 \Rightarrow P = 500$ thousand. At $P = 400$: $\dfrac{dP}{dt} = 0.04(400) - 20 = -4 < 0$, so P is decreasing.

Topic 7.2 check.

1. $\dfrac{dy}{dx} = 3e^{3x} + 1$. Right side: $3y - 3x + 1 = 3(e^{3x} + x) - 3x + 1 = 3e^{3x} + 1$. The sides agree, so y is a solution.
2. $y' = ke^{kx}$, $y'' = k^2e^{kx}$. Then $(k^2 - 5k + 6)e^{kx} = 0$ and $e^{kx} \neq 0$, so $(k-2)(k-3) = 0$: $k = 2$ or $k = 3$.
3. $\dfrac{dy}{dx} = \dfrac{1}{(1-x)^2}$ and $y^2 = \dfrac{1}{(1-x)^2}$: equal. Also $y(0) = \dfrac{1}{1} = 1$.

Topic 7.3 check.

1. $(0,0)$: 0. $(1,-1)$: 0. $(-2,1)$: -1. $(2,3)$: 5.
2. The right side depends only on y and is 0 at $y = 2$: segments are horizontal along $y = 2$; rows are identical. For $y < 2$ the slope $2 - y > 0$, so segments rise; for $y > 2$ they fall.
3. The right side $\frac{x}{2}$ contains no y, so the slope is the same at every point with a given x (a vertical column). At $x = -2$ the slope is -1.

Topic 7.4 check.

1. $\dfrac{dy}{dx} = y - 1$ is 0 exactly when $y = 1$ (for every x); $x - 1$ is 0 on $x = 1$ instead. For $\dfrac{dy}{dx} = y - 1$: $\dfrac{d^2y}{dx^2} = \dfrac{dy}{dx} = y - 1$.
2. $\dfrac{d^2y}{dx^2} = 1 - \dfrac{dy}{dx} = 1 - (x - y)$. At $(1,1)$: $\dfrac{dy}{dx} = 0$, so $\dfrac{d^2y}{dx^2} = 1 > 0$: concave up (and f has a relative minimum at $x = 1$).
3. At $(0,5)$ the slope is $2 - 5 = -3 < 0$, and for $y > 2$ the slope stays negative, so f is decreasing. The line $y = 2$ is an equilibrium solution that f cannot cross, so f is decreasing and bounded below by 2; $\lim\limits_{x \to \infty} f(x) = 2$.

Topic 7.6 check.

1. $\dfrac{dy}{y} = 3x^2\,dx \Rightarrow \ln|y| = x^3 + C \Rightarrow y = Ae^{x^3}$.
2. $y^2\,dy = e^x\,dx \Rightarrow \dfrac{y^3}{3} = e^x + C \Rightarrow y^3 = 3e^x + K$, i.e. $y = \sqrt[3]{3e^x + K}$.
3. $\dfrac{dy}{y+1} = \dfrac{dx}{x} \Rightarrow \ln|y+1| = \ln x + C \Rightarrow y + 1 = Ax \Rightarrow y = Ax - 1$.

Topic 7.7 check.

1. $y\,dy = -x\,dx \Rightarrow \dfrac{y^2}{2} = -\dfrac{x^2}{2} + C$; $y(0) = 2$ gives $C = 2$, so $x^2 + y^2 = 4$. Since $y(0) > 0$: $y = \sqrt{4 - x^2}$, domain $-2 < x < 2$ (the derivative $-x/y$ is undefined at $y = 0$, i.e. $x = \pm 2$).
2. $\dfrac{dy}{y^2} = e^x\,dx \Rightarrow -\dfrac{1}{y} = e^x + C$; $y(0) = 1$: $-1 = 1 + C$, $C = -2$. So $-\dfrac{1}{y} = e^x - 2$, $y = \dfrac{1}{2 - e^x}$, defined for $x < \ln 2$.
3. $\dfrac{dy}{y} = 3\,dx \Rightarrow \ln|y| = 3x + C$; $\ln 5 = 3 + C$; $y = 5e^{3x-3} = 5e^{3(x-1)}$.

Topic 7.8 check.

1. $P(t) = 250e^{0.03t}$, so $P(10) = 250e^{0.3} \approx 337.465$.
2. $e^{kt} = \frac{1}{8}$ with $k = -\dfrac{\ln 2}{12}$: $t = \dfrac{\ln(1/8)}{k} = \dfrac{-3\ln 2}{-\ln 2/12} = 36$ days (three half-lives).
3. $T = 10 + (30 - 10)e^{-0.2t} = 10 + 20e^{-0.2t}$, so $T(5) = 10 + 20e^{-1} \approx 17.358$.

Part A – Multiple Choice, No Calculator

Solutions 1–15

1. Answer: C. (easy)

1. "Proportional to the square root of P" means rate $= k \cdot \sqrt{P}$: $\dfrac{dP}{dt} = k\sqrt{P}$.

Why not the others: (A) is *inversely* proportional to $\sqrt{P}$. (B) is proportional to P^2 (wrong function). (D) *adds* k instead of multiplying ("proportional" misread as "plus a constant").

2. Answer: B. (easy)

1. For $y = x + 1 + 2e^x$: $\dfrac{dy}{dx} = 1 + 2e^x$. Right side: $y - x = 1 + 2e^x$. Equal.
2. Initial condition: $y(0) = 0 + 1 + 2 = 3$. ✓

Why not the others: (A) $y(0) = 3$ but $y' = e^x \neq y - x = e^x + 2 - x$ (checks only the initial condition). (C) $y(0) = 3$ but $y' = 1 + 3e^x \neq y - x = 3e^x$. (D) $y(0) = 3$ but $y' = 3e^x \neq 3e^x - x$.

3. Answer: A. (easy)

1. Substitute $x = -1$, $y = 3$: $\dfrac{dy}{dx} = (-1)^2 - 2(3) = 1 - 6 = -5$.

Why not the others: (B) uses $-x^2$ $(-1 - 6)$: $(-1)^2$ evaluated as -1. (C) sign error on the y-term $(1 + 6)$. (D) swaps the coordinates, $x = 3$, $y = -1$: $9 + 2$.

4. Answer: D. (easy)

1. Read the segment at $(0, 1)$: it falls with slope -1. Test each equation at $(0, 1)$: $x - y = -1$ ✓; $x + y = 1$; $y - x = 1$; $xy = 0$.
2. Check a second point: horizontal segments lie on the line $y = x$ (e.g. $(2, 2)$), where $x - y = 0$. ✓

Why not the others: (A) $x + y$ has horizontal segments on $y = -x$. (B) $y - x$ is the negative of the correct equation (rising where the field falls). (C) xy is horizontal along both axes, but the field is not horizontal at $(0, 1)$.

5. Answer: C. (medium)

1. Differentiate with respect to x: $\dfrac{d^2y}{dx^2} = \dfrac{dy}{dx} - 1$.
2. At $(1, 3)$: $\dfrac{dy}{dx} = 3 - 1 = 2$, so $\dfrac{d^2y}{dx^2} = 2 - 1 = 1$.

Why not the others: (A) gives $\dfrac{dy}{dx}$ instead of $\dfrac{d^2y}{dx^2}$. (B) gives the y-value. (D) treats y as a constant when differentiating ($\frac{d}{dx}(y - x) = -1$), forgetting that $\frac{dy}{dx} \neq 0$.

6. Answer: A. (medium)

1. Separate: $\dfrac{dy}{y} = x^2\,dx$.
2. Integrate: $\ln|y| = \dfrac{x^3}{3} + C$.
3. $y = Ae^{x^3/3}$ (absorbing $\pm e^C$ into A).

Why not the others: (B) ignores the y on the right side (integrates x^2 only). (C) adds the constant after exponentiating. (D) integrates x^2 as x^3 (loses the $\frac{1}{3}$).

7. Answer: D. (medium)

1. $y\,dy = 4x\,dx \Rightarrow \dfrac{y^2}{2} = 2x^2 + C$.
2. $(1, -3)$: $\dfrac{9}{2} = 2 + C$, $C = \dfrac{5}{2}$; $y^2 = 4x^2 + 5$.
3. $y(1) < 0$, so $y = -\sqrt{4x^2 + 5}$.

Why not the others: (A) takes the positive root, ignoring $y(1) = -3$. (B) sets $C = y(1)^2 = 9$ without subtracting the x-term ($4x^2$ at $x = 1$), so $y(1) = -\sqrt{13} \neq -3$. (C) integrates $4x$ as $2x$ (forgetting the factor 2 from $\frac{y^2}{2}$), giving $y^2 = 2x^2 + 7$; this matches $y(1) = -3$ but does not satisfy the differential equation.

8. Answer: B. (easy)

1. $\dfrac{dy}{dt} = \frac{1}{2}y$ gives $y = 6e^{t/2}$.

2. $y(2) = 6e^1 = 6e$.

Why not the others: (A) uses exponent $t = 2$ without the factor $\frac{1}{2}$. (C) is linear growth: $6 + \frac{1}{2}(6)(2) = 12$. (D) adds e to 6 instead of multiplying.

9. Answer: A. (medium)

1. The right side is 0 at $y = \pm 2$, so $y = -2$ and $y = 2$ are equilibrium solutions.
2. At $y = 1$: $\dfrac{dy}{dx} = \dfrac{1-4}{4} = -\dfrac{3}{4} < 0$; for $-2 < y < 2$ the slope is negative, so f is decreasing.
3. f cannot cross $y = -2$, so it is decreasing and bounded below by -2 and the segments flatten there: $\lim\limits_{x\to\infty} f(x) = -2$.

Why not the others: (B) takes the midpoint of the two equilibria. (C) picks the wrong equilibrium: solutions near $y = 2$ move *away* from it. (D) assumes "decreasing" means unbounded, ignoring the barrier at $y = -2$.

10. Answer: B. (hard)

1. $\dfrac{dy}{dx} = xy$, so $f'(0) = 0 \cdot 1 = 0$: a critical point.
2. $\dfrac{d^2y}{dx^2} = y + x\dfrac{dy}{dx}$. At $(0, 1)$: $1 + 0 = 1 > 0$.
3. Since $f'(0) = 0$ and $f''(0) > 0$, f has a relative minimum at $x = 0$.

Why not the others: (A) reverses the sign of the second-derivative test. (C) ignores that $f'(0) = 0$ makes $x = 0$ a critical point (and a minimum). (D) an inflection needs $f''(0) = 0$ with a sign change, but $f''(0) = 1$.

11. Answer: C. (medium)

1. $\dfrac{dT}{T-10} = -\frac{1}{2}\,dt \Rightarrow T - 10 = Ae^{-t/2}$; $T(0) = 30$: $A = 20$.
2. $T(t) = 10 + 20e^{-t/2}$, so $T(2) = 10 + 20e^{-1}$.

Why not the others: (A) uses $T = T_0e^{kt}$, ignoring the surroundings temperature 10. (B) drops the $+10$. (D) uses $t = 1$ (exponent $-\frac{1}{2}$).

12. Answer: D. (medium)

1. The equation $\dfrac{dA}{dt} = -\dfrac{1}{50}(A - 150)$ has equilibrium $A = 150$; solving, $A = 150(1 - e^{-t/50})$.
2. As $t \to \infty$, $e^{-t/50} \to 0$ and $A \to 150$.

Why not the others: (A) is the inflow rate (kg per minute), not an amount. (B) is the 50 in the denominator. (C) is the initial amount, not the limit.

13. Answer: C. (hard)

1. $y^{-2}\,dy = \cos x\,dx \Rightarrow -\dfrac{1}{y} = \sin x + C$.
2. $y(0) = 1$: $-1 = 0 + C$, so $-\dfrac{1}{y} = \sin x - 1$ and $y = \dfrac{1}{1 - \sin x}$.
3. $y\left(\frac{\pi}{6}\right) = \dfrac{1}{1 - \frac{1}{2}} = 2$.

Why not the others: (A) loses the sign in $\int y^{-2}dy = -y^{-1}$ giving $y = \dfrac{1}{\sin x + 1}$. (B) integrates y^{-2} as $\ln|y|$, giving $y = e^{\sin x}$. (D) takes $C = +1$ ($-1/y = \sin x + 1$), a sign error when using the initial condition.

14. Answer: B. (medium)

1. For $y = e^{-3x}$: $y' = -3y$, $y'' = 9y$.
2. (B): $y'' + y' - 6y = 9y - 3y - 6y = 0$. ✓

Why not the others: (A) $9y + 3y - 6y = 6y$: this equation is satisfied by $y = e^{3x}$ ($k = 3$, sign error on k). (C) $9y - 3y + 6y = 12y$ (sign error on the constant term). (D) $9y - 6y = 3y$ (drops the y' term).

15. Answer: A. (medium)

1. $\dfrac{dy}{y} = x^{-2}\,dx \Rightarrow \ln|y| = -x^{-1} + C = -\dfrac{1}{x} + C$.
2. $y = Ce^{-1/x}$.

Why not the others: (B) integrates x^{-2} as x^{-1} (power-rule sign error). (C) forgets the y in the denominator

(integrates only x^{-2}). (D) differentiates x^{-2} instead of integrating it, giving $-2x^{-3}$.

Part B – Multiple Choice, Graphing Calculator Required

Solutions 16–22

16. Answer: A. (medium)

1. $T - 20 = 75e^{-0.08t}$. Set $T = 50$: $75e^{-0.08t} = 30 \Rightarrow e^{-0.08t} = 0.4$.
2. $t = \dfrac{\ln 0.4}{-0.08} = \dfrac{\ln 2.5}{0.08} \approx 11.454$.

Why not the others: (B) forgets to subtract 20: $95e^{-0.08t} = 50$. (C) uses $\dfrac{50}{75}$ (subtracts 20 on one side only). (D) divides 0.4 by 0.08 without taking the logarithm.

17. Answer: B. (easy)

1. Doubling every 8 hours: $k = \dfrac{\ln 2}{8}$. Solve $e^{kt} = 5$: $t = \dfrac{\ln 5}{k} = \dfrac{8 \ln 5}{\ln 2} \approx 18.575$.

Why not the others: (A) $8 \ln 5$ (forgets to divide by $\ln 2$). (C) 5×8 (treats growth as linear). (D) $\dfrac{\ln 5}{\ln 2}$ (forgets the factor 8).

18. Answer: D. (medium)

1. Separate: $e^y \, dy = x \, dx$; integrate: $e^y = \dfrac{x^2}{2} + C$.
2. $y(0) = 0$: $1 = C$. So $y = \ln\left(\dfrac{x^2}{2} + 1\right)$.
3. $y(2) = \ln(2 + 1) = \ln 3 \approx 1.099$.

Why not the others: (A) omits the constant ($C = 0$): $\ln 2$. (B) integrates x as x^2 (forgets the $\frac{1}{2}$): $\ln(x^2+1) = \ln 5$. (C) ignores the e^{-y} factor: $y = \dfrac{x^2}{2} = 2$.

19. Answer: C. (medium)

1. $\dfrac{dQ}{dt} = -0.1(Q - 50)$, so $Q = 50 + (0 - 50)e^{-0.1t} = 50\left(1 - e^{-0.1t}\right)$.
2. $Q(8) = 50(1 - e^{-0.8}) \approx 50(0.5507) = 27.534$ mg.

Why not the others: (A) 5×8 ignores elimination. (B) $50e^{-0.8}$ forgets the factor $1 - \ldots$ (it is the amount *missing* from 50). (D) is the equilibrium amount, reached only as $t \to \infty$.

20. Answer: D. (hard)

1. $\dfrac{dy}{1 + y^2} = dx \Rightarrow \arctan y = x + C$; $y(0) = 0$: $C = 0$.
2. $y = \tan x$, so $y(1) = \tan 1 \approx 1.557$.

Why not the others: (A) writes $y = \arctan x$ (inverts the relation). (B) assumes $y = x$ because $\dfrac{dy}{dx} = 1$ at $(0, 0)$ (extends the tangent line). (C) separates incorrectly as $(1 + y^2)\, dy = dx$, giving $y + \dfrac{y^3}{3} = x$, whose solution at $x = 1$ is ≈ 0.818.

21. Answer: B. (easy)

1. $P = 250e^{0.03t}$ and $\dfrac{dP}{dt} = 0.03P$.
2. At $t = 10$: $\dfrac{dP}{dt} = 0.03 \cdot 250e^{0.3} \approx 0.03(337.465) = 10.124$ thousand per year.

Why not the others: (A) is $P(10)$, the population, not its rate. (C) is the rate at $t = 0$ $(0.03 \cdot 250)$. (D) is $kP_0t = 0.03 \cdot 250 \cdot 10$, treating growth as linear.

22. Answer: D. (hard)

1. $T - 20 = 60e^{kt}$. At $t = 5$: $60e^{5k} = 30 \Rightarrow e^{5k} = \frac{1}{2}$.
2. $k = -\dfrac{\ln 2}{5} \approx -0.139$.

Why not the others: (A) has the wrong sign (cooling needs $k < 0$). (B) uses $\ln \dfrac{50}{80}$, forgetting to subtract 20 from

both temperatures. (C) uses the average rate, $\dfrac{-30/5}{60} = -0.1$, treating the cooling as linear.

Part C – Free Response (scoring guide)

Solutions 23–24

23. (9 points)

(a) $\left.\dfrac{dT}{dt}\right|_{t=0} = -\dfrac{1}{5}(90-22) = -13.6.$

1 point: tangent line $y = 90 - 13.6\,t$ (or $T - 90 = -13.6\,t$).

1 point: $T(1) \approx 90 - 13.6 = 76.4°\text{C}$.

(b) $\dfrac{d^2T}{dt^2} = -\dfrac{1}{5}\dfrac{dT}{dt} = -\dfrac{1}{5}\left(-\dfrac{1}{5}(T-22)\right) = \dfrac{1}{25}(T-22).$

1 point: the second derivative.

1 point: For $t \geq 0$, $T > 22$, so $\dfrac{d^2T}{dt^2} > 0$ and the graph is concave up; the tangent line lies below the graph, so 76.4 is an *underestimate*. (Check: $T(1) = 22 + 68e^{-0.2} \approx 77.674$.)

(c) **1 point:** separation $\dfrac{dT}{T-22} = -\dfrac{1}{5}\,dt.$

1 point: antiderivatives, with the constant: $\ln|T-22| = -\dfrac{t}{5} + C.$

1 point: use $T(0) = 90$: $\ln 68 = C$.

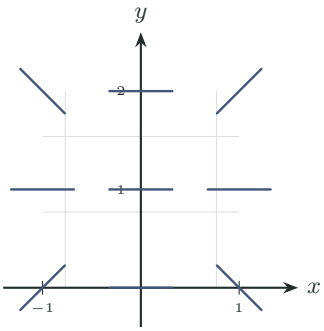
1 point: $|T-22| = 68e^{-t/5}$ and, since $T(0) - 22 > 0$, $T(t) = 22 + 68e^{-t/5}$.

(d) **1 point:** $22 + 68e^{-t/5} = 40 \Rightarrow e^{-t/5} = \dfrac{18}{68} \Rightarrow t = 5\ln\dfrac{68}{18} \approx 6.646$ minutes.

24. (9 points)

(a) Slopes $x(y-1)$:

$y\backslash x$	-1	0	1
2	-1	0	1
1	0	0	0
0	1	0	-1



1 point: horizontal segments at the five points where the slope is 0 (the whole row $y = 1$ and the whole column $x = 0$).

1 point: the four remaining segments with the correct signs: slope 1 at $(-1,0)$ and $(1,2)$, slope -1 at $(-1,2)$ and $(1,0)$.

(b) $\dfrac{d^2y}{dx^2} = \dfrac{d}{dx}\left[x(y-1)\right] = (y-1) + x\dfrac{dy}{dx} = (y-1) + x^2(y-1) = (1+x^2)(y-1).$

1 point: the second derivative.

At $(0,3)$: $\dfrac{dy}{dx} = 0 \cdot 2 = 0$ and $\dfrac{d^2y}{dx^2} = (1)(2) = 2 > 0$. **1 point:** since $f'(0) = 0$ and $f''(0) > 0$, f has a relative minimum at $x = 0$.

(c) **1 point:** $\dfrac{dy}{y-1} = x\,dx.$

1 point: $\ln|y-1| = \dfrac{x^2}{2} + C.$

1 point: $f(0) = 3$: $\ln 2 = C$.

1 point: $|y-1| = 2e^{x^2/2}$ and since $y(0) - 1 > 0$: $y = 1 + 2e^{x^2/2}$.

(d) **1 point:** $\lim\limits_{x\to\infty} f(x) = \infty$ (since $e^{x^2/2} \to \infty$).

Unit 8 – Worked Solutions

Topic 8.1 check

1. $\frac{1}{2}\int_0^2 x^3 dx = \frac{1}{2}\cdot\frac{2^4}{4} = 2.$ **2.** $\frac{1}{5-1}\cdot 14 = \frac{7}{2}.$

3. $f_{\text{avg}} = \frac{1}{\pi}\int_0^\pi \sin x\,dx = \frac{2}{\pi} \approx 0.6366$. Solve $\sin c = \frac{2}{\pi}$ on $[0,\pi]$: $c = \arcsin\frac{2}{\pi} \approx 0.690$ and $c = \pi - 0.690 \approx 2.452$.

Topic 8.2 check

1. $V(t) = t^3 - 12t$. Displacement $= V(3) - V(0) = -9$. $v = 0$ at $t = 2$; $V(2) = -16$, so $\int_0^2 v = -16$ and $\int_2^3 v = -9 - (-16) = 7$. Total distance $= 16 + 7 = 23$.

2. $v(t) = 2t^2 + C$, $v(0) = -2$, so $v(t) = 2t^2 - 2$. $x(3) = 3 + \int_0^3 (2t^2 - 2)\,dt = 3 + \left[\frac{2}{3}t^3 - 2t\right]_0^3 = 3 + (18 - 6) = 15$.

3. $v(1) = 1 - 4 = -3$ and $a(t) = 2t - 4$, $a(1) = -2$. Both are negative, so the speed is **increasing** at $t = 1$.

Topic 8.3 check

1. $80 - \int_0^6 (12 - 2t)\,dt = 80 - \left[12t - t^2\right]_0^6 = 80 - 36 = 44$ liters.

2. $\int_0^{10} R(t)\,dt$ is the total number of people who enter the stadium during the first 10 minutes (units: people).

3. $R(t) = (t-2)(t-4)$. $R > 0$ on $(0,2)$ and $R < 0$ on $(2,4)$, so the amount changes from increasing to decreasing at $t = 2$: a local maximum at $t = 2$.

Topic 8.4 check

1. $x^2 = 8 - x^2 \Rightarrow x = \pm 2$. $A = \int_{-2}^2 (8 - 2x^2)\,dx = 2\left[8x - \frac{2}{3}x^3\right]_0^2 = 2\left(16 - \frac{16}{3}\right) = \frac{64}{3}$.

2. $x + 1 = x^2 - 1 \Rightarrow (x-2)(x+1) = 0$. $A = \int_{-1}^2 (-x^2 + x + 2)\,dx = \left[-\frac{x^3}{3} + \frac{x^2}{2} + 2x\right]_{-1}^2 = \frac{10}{3} - \left(-\frac{7}{6}\right) = \frac{9}{2}$.

Topic 8.5 check

1. Meets at $y = \pm 2$; $A = \int_{-2}^2 (4 - y^2)\,dy = 2\left(8 - \frac{8}{3}\right) = \frac{32}{3}$.

2. $A = \int_0^1 e^y\,dy = e - 1 \approx 1.718$.

Topic 8.6 check

1. $x^3 - 4x = x(x-2)(x+2)$, zeros $-2, 0, 2$. By symmetry $A = 2\int_0^2 (4x - x^3)\,dx = 2\left[2x^2 - \frac{x^4}{4}\right]_0^2 = 2(8-4) = 8$.

2. $x^3 = 2x \Rightarrow x = 0, \pm\sqrt{2}$; $A = 2\int_0^{\sqrt{2}} (2x - x^3)\,dx = 2\left[x^2 - \frac{x^4}{4}\right]_0^{\sqrt{2}} = 2(2-1) = 2$.

Topic 8.7 check

1. $s = x$, $A = x^2$: $V = \int_0^3 x^2 dx = 9$. **2.** $s = x - x^2$ on $[0,1]$: $V = 5\int_0^1 (x - x^2)\,dx = 5\cdot\frac{1}{6} = \frac{5}{6}$.

3. Areas w^2: $1, 9, 16, 4$. $V \approx \frac{1}{2}(1)(1+9) + \frac{1}{2}(2)(9+16) + \frac{1}{2}(1)(16+4) = 5 + 25 + 10 = 40$.

Topic 8.8 check

1. $d = \sqrt{x}$, $A = \frac{\pi}{8}x$: $V = \dfrac{\pi}{8}\displaystyle\int_0^4 x\,dx = \frac{\pi}{8}\cdot 8 = \pi$.

2. $s = x - x^2$: $V = \dfrac{\sqrt{3}}{4}\displaystyle\int_0^1 (x-x^2)^2 dx = \frac{\sqrt{3}}{4}\cdot\frac{1}{30} = \frac{\sqrt{3}}{120}$.

3. Meets where $y^2 = y+2$: $y = -1, 2$. $d = y + 2 - y^2$. $V = \dfrac{\pi}{8}\displaystyle\int_{-1}^2 (y+2-y^2)^2 dy = \frac{\pi}{8}\cdot\frac{81}{10} = \frac{81\pi}{80} \approx 3.181$.

Topic 8.9 check

1. $\pi\displaystyle\int_0^3 x^2 dx = 9\pi$. **2.** $x = y^{1/3}$: $\pi\displaystyle\int_0^8 y^{2/3} dy = \pi\cdot\tfrac{3}{5}\cdot 8^{5/3} = \frac{96\pi}{5}$.

3. $\pi\displaystyle\int_0^{\pi} \sin^2 x\,dx = \pi\cdot\frac{\pi}{2} = \frac{\pi^2}{2} \approx 4.935$.

Topic 8.10 check

1. $R = 1 - x^2$, $0 \le x \le 1$: $\pi\displaystyle\int_0^1 (1-x^2)^2 dx = \pi\left(1 - \tfrac{2}{3} + \tfrac{1}{5}\right) = \frac{8\pi}{15}$.

2. Slice horizontally: $x = \sqrt{y}$, $0 \le y \le 1$, $R = 1 - \sqrt{y}$: $\pi\displaystyle\int_0^1 (1 - 2\sqrt{y} + y)\,dy = \pi\left(1 - \tfrac{4}{3} + \tfrac{1}{2}\right) = \frac{\pi}{6}$.

Topic 8.11 check

1. $R = x$, $r = x^3$: $\pi\displaystyle\int_0^1 (x^2 - x^6)\,dx = \pi\left(\tfrac{1}{3} - \tfrac{1}{7}\right) = \frac{4\pi}{21}$.

2. $x = y^2$ and $x = y$; for $0 \le y \le 1$, $y \ge y^2$: $R = y$, $r = y^2$. $V = \pi\displaystyle\int_0^1 (y^2 - y^4)\,dy = \frac{2\pi}{15}$.

3. Meets at $x = \pm 2$; $R = 4$, $r = x^2$: $\pi\displaystyle\int_{-2}^2 (16 - x^4)\,dx = 2\pi\left(32 - \tfrac{32}{5}\right) = \frac{256\pi}{5}$.

Topic 8.12 check

1. Axis below the region: $R = x + 1$ (upper curve $y = x$), $r = x^2 + 1$. $V = \pi\displaystyle\int_0^1 \left[(x+1)^2 - (x^2+1)^2\right]dx = \pi\displaystyle\int_0^1 (2x - x^2 - x^4)\,dx = \pi\left(1 - \tfrac{1}{3} - \tfrac{1}{5}\right) = \frac{7\pi}{15}$.

2. Axis to the left; for $0 \le y \le 1$ the right boundary is $x = y$ and the left boundary is $x = y^2$. $R = y + 2$, $r = y^2 + 2$. $V = \pi\displaystyle\int_0^1 \left[(y+2)^2 - (y^2+2)^2\right]dy = \pi\displaystyle\int_0^1 (4y - 3y^2 - y^4)\,dy = \pi\left(2 - 1 - \tfrac{1}{5}\right) = \frac{4\pi}{5}$.

Part A – Solutions

No calculator

1. Answer: (B). 1. $\int_1^3 (3x^2 - 2x)\,dx = \left[x^3 - x^2\right]_1^3 = 18 - 0 = 18$. 2. $f_{\text{avg}} = \frac{18}{3-1} = 9$.
Why not the others: (A) 10 is the average *rate of change*, $\frac{f(3)-f(1)}{2}$. (C) 11 averages the endpoint values $\frac{f(1)+f(3)}{2}$. (D) 18 is the integral; the factor $\frac{1}{b-a}$ was forgotten.

2. Answer: (D). 1. $v = 3t(t-2)$, so $v < 0$ on $(0,2)$ and $v > 0$ on $(2,3)$. 2. With $V(t) = t^3 - 3t^2$: $V(2) - V(0) = -4$, $V(3) - V(2) = 0 - (-4) = 4$. 3. Distance $= |-4| + |4| = 8$.
Why not the others: (A) -4 is the displacement on $[0,2]$ only (and a distance cannot be negative). (B) 0 is the displacement on $[0,3]$ (no splitting at the turn). (C) 4 is the distance on $[0,2]$ only; the motion after $t = 2$ was ignored.

3. Answer: (A). 1. $x^2 = x + 2 \Rightarrow x = -1, 2$; the line is on top. 2. $A = \int_{-1}^2 (x + 2 - x^2)\,dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3}\right]_{-1}^2 =$

$\frac{10}{3}+\frac{7}{6}=\frac{9}{2}$.
Why not the others: (B) $\frac{10}{3}$ uses 0 instead of -1 as the lower limit. (C) $-\frac{9}{2}$ subtracts bottom − top. (D) $\frac{21}{2}$ adds the curves instead of subtracting.

4. Answer: (C). $V=\pi\int_0^2(x^2)^2dx=\pi\int_0^2 x^4dx=\pi\cdot\frac{32}{5}=\frac{32\pi}{5}$.
Why not the others: (A) $\frac{8\pi}{3}$ forgets to square the radius ($\pi\int x^2$). (B) 32π integrates x^4 as x^5 (no division by 5). (D) $\frac{32}{5}$ omits the factor π.

5. Answer: (B). 1. Curves meet at $x=0,1$; $R=x$, $r=x^2$. 2. $V=\pi\int_0^1(x^2-x^4)\,dx=\pi\left(\frac{1}{3}-\frac{1}{5}\right)=\frac{2\pi}{15}$.
Why not the others: (A) $\frac{\pi}{30}=\pi\int(R-r)^2$ (subtracting before squaring). (C) $\frac{\pi}{6}=\pi\int(R-r)$ (radii not squared). (D) $\frac{\pi}{3}$ uses only the outer radius (the hole ignored).

6. Answer: (D). 1. Slice at height y: x from $-\sqrt{y}$ to $\sqrt{y}$, so $s=2\sqrt{y}$. 2. $A=s^2=4y$. 3. $V=\int_0^4 4y\,dy=2y^2\big|_0^4=32$.
Why not the others: (A) 8 uses $s=\sqrt{y}$ (half the width). (B) $\frac{32}{3}$ integrates the side instead of its square. (C) $\frac{512}{15}$ is the volume for squares perpendicular to the x-axis, $\int_{-2}^2(4-x^2)^2dx$.

7. Answer: (C). $\frac{1}{2}(3)(4+6)+\frac{1}{2}(2)(6+5)+\frac{1}{2}(4)(5+7)=15+11+24=50$.
Why not the others: (A) 44 is the left sum, (B) 56 the right sum, (D) 100 omits the factor $\frac{1}{2}$.

8. Answer: (A). The integral of a rate over $[2,5]$ is the accumulated change: the total number of tickets sold from hour 2 to hour 5.
Why not the others: (B) would be $\frac{1}{3}\int_2^5 R$ (an average rate). (C) The integral is a change over an interval, not an amount at one instant. (D) $R(5)-R(2)$ is $\int_2^5 R'(t)\,dt$.

9. Answer: (C). $\frac{1}{3}\int_1^4\big(3f(x)+2\big)dx=\frac{1}{3}\big(3\cdot12+2\cdot3\big)=\frac{42}{3}=14$.
Why not the others: (A) $\frac{38}{3}$ integrates the constant 2 over a length of 1 instead of 3. (B) 12 forgets the $+2$. (D) 38 forgets to divide by $b-a$.

10. Answer: (B). 1. Use y: $x=y^2$ (left) and $x=y+2$ (right); they meet at $y=2$. The x-axis is $y=0$. 2. $A=\int_0^2(y+2-y^2)\,dy=2+4-\frac{8}{3}=\frac{10}{3}$.
Why not the others: (A) $\frac{16}{3}=\int_0^4\big(\sqrt{x}-(x-2)\big)dx$ includes the part of the triangle below the x-axis. (C) $\frac{8}{3}=\int_0^2 y^2dy$ uses only the left curve. (D) $\frac{26}{3}$ adds y^2 instead of subtracting.

11. Answer: (D). 1. The axis $y=-1$ is below the region, so $R=\sqrt{x}+1$ and $r=0+1=1$. 2. $V=\pi\int_0^4\left[(\sqrt{x}+1)^2-1\right]dx=\pi\int_0^4(x+2\sqrt{x})\,dx=\pi\left(8+\frac{32}{3}\right)=\frac{56\pi}{3}$.
Why not the others: (A) $\frac{68\pi}{3}$ drops the hole ($r=1$). (B) $\frac{4\pi}{3}=\pi\int(\sqrt{x}-1)^2$ treats the axis as $y=1$. (C) 8π is the volume about the x-axis.

12. Answer: (A). Areas: $[0,2]$ triangle 4; $[2,4]$ rectangle 8; $[4,5]$ triangle 2 (above); $[5,6]$ triangle 2 (below); $[6,8]$ triangle 4 (below). The segment from $(4,4)$ to $(6,-4)$ crosses zero at $t=5$. Distance $=4+8+2+2+4=20$.
Why not the others: (B) 14 counts only motion to the right. (C) 6 counts only motion to the left. (D) 8 is the displacement $14-6$.

13. Answer: (B). $v(t)=3t^2-12t+9=3(t-1)(t-3)$, $a(t)=6t-12$. The speed is increasing when v and a have the same sign. $t=1.5$: $v=-2.25<0$, $a=-3<0$: increasing.
Why not the others: $t=0.5$: $v=3.75>0$, $a=-9<0$ (slowing down). $t=2$: $a=0$, speed momentarily unchanged. $t=2.5$: $v=-2.25<0$, $a=3>0$ (slowing down).

14. Answer: (C). 1. $W'(t)=(t-1)(t-4)$: W increases on $(0,1)$, decreases on $(1,4)$, increases on $(4,6)$. 2. Candidates: $W(0)=0$; $W(1)=\frac{1}{3}-\frac{5}{2}+4=\frac{11}{6}$; $W(4)=\frac{64}{3}-40+16=-\frac{8}{3}$; $W(6)=72-90+24=6$. The greatest is $W(6)=6$.
Why not the others: (A) $t=0$ gives 0. (B) $t=1$ is a local maximum only (the endpoint was not checked). (D) $t=4$ is a local minimum.

Part B – Solutions

Graphing calculator

15. Answer: (A). $\frac{1}{2}\int_0^2 e^{-x^2}dx = \frac{1}{2}(0.88208) \approx 0.441$.
Why not the others: (B) 0.882 is the integral (no division by 2). (C) $-0.491 = \frac{f(2)-f(0)}{2}$ is the average rate of change. (D) $0.509 = \frac{f(0)+f(2)}{2}$ averages the endpoints.

16. Answer: (D). 1. Solve $e^x = 4 - x^2$: $x \approx -1.965$ and $x \approx 1.058$. At $x = 0$: $g = 4 > f = 1$, so g is on top. 2. $A = \int_{-1.965}^{1.058}(4 - x^2 - e^x)dx \approx 6.428$.
Why not the others: (A) 1.957 uses 0 as the lower limit. (B) 9.442 uses $[-2, 2]$ (guessed limits, not the intersections). (C) -6.428 subtracts $f - g$ (bottom − top).

17. Answer: (B). $V = \pi\int_1^e (\ln x)^2 dx = \pi\left[x(\ln x)^2 - 2x\ln x + 2x\right]_1^e = \pi(e-2) \approx 2.257$.
Why not the others: (A) $0.718 = e - 2$ omits π. (C) $3.142 = \pi\int_1^e \ln x\,dx$ does not square the radius. (D) $4.513 = 2\pi(e-2)$ doubles the volume.

18. Answer: (C). $v(t) = \sin(t^2)$ changes sign at $t = \sqrt{\pi} \approx 1.772$. Distance $= \int_0^2 |\sin(t^2)|\,dt \approx 0.985$.
Why not the others: (A) $0.804 = \int_0^2 v\,dt$ is the displacement. (B) $-0.757 = v(2)$. (D) $0.895 = \int_0^{\sqrt{\pi}} v\,dt$ stops at the turning point.

19. Answer: (A). $80 - \int_0^5 10e^{-0.3t}dt = 80 - 25.896 = 54.104$.
Why not the others: (B) 25.896 is the amount that leaked, not what remains. (C) 105.896 adds the outflow. (D) $77.769 = 80 - r(5)$ uses the rate instead of the integral.

20. Answer: (D). $s = \ln x$, $A = (\ln x)^2$; $V = \int_1^3 (\ln x)^2 dx \approx 1.029$.
Why not the others: (A) $1.296 = \int_1^3 \ln x\,dx$ (side not squared). (B) $1.679 = \left(\int \ln x\right)^2$ (squared after integrating). (C) $3.233 = \pi\int(\ln x)^2$ (discs instead of squares).

21. Answer: (B). 1. $x^2 = 3 - x \Rightarrow x = \frac{-1\pm\sqrt{13}}{2} \approx -2.303,\ 1.303$. Both curves lie above the x-axis, so $R = 3 - x$, $r = x^2$. 2. $V = \pi\int_{-2.303}^{1.303}\left[(3-x)^2 - x^4\right]dx \approx 107.986$.
Why not the others: (A) 34.373 omits π. (C) $63.810 = \pi\int(R-r)^2$. (D) 151.029 uses only the outer radius.

Part C – Solutions (scoring guide style)

22.

(a) 2 points.
1 point: $\displaystyle\int_0^{10} R(t)\,dt = \int_0^{10}\left(12 + 8\cos\frac{t}{3}\right)dt$.
1 point: 115.426 gallons.

(b) 2 points.
1 point: $R(7) - D(7) \approx 6.474 - 10 = -3.526$.
1 point: The rate of change of the amount, $R(7) - D(7)$, is negative, so the amount of water is **decreasing** at $t = 7$.

(c) 3 points.
1 point: $A'(t) = R(t) - D(t) = 0$, i.e. $9 + 8\cos\frac{t}{3} - t = 0$.
1 point: $t \approx 5.904$ minutes.
1 point: Justification: $A'(t) = R(t) - D(t)$ is positive for $0 \le t < 5.904$ (for instance $A'(5) \approx 3.234$) and negative for $5.904 < t \le 10$ ($A'(6) \approx -0.329$); it is strictly decreasing since $\frac{d}{dt}(R - D) = -\frac{8}{3}\sin\frac{t}{3} - 1 < 0$. So the amount increases and then decreases and is greatest at $t = 5.904$. (Equivalent: compare $A(0) = 50$, $A(5.904) = 107.839$, $A(10) = 85.426$.)

(d) 2 points.
1 point: $A(5.904) = 50 + \displaystyle\int_0^{5.904}\big(R(t) - D(t)\big)dt$.
1 point: 107.839 gallons.

23.

(a) 2 points.

1 point: $f(x) = g(x)$ gives $3x - x^2 = x$, $x = 0, 2$; $A = \displaystyle\int_0^2 \left[(3x - x^2) - x\right] dx = \int_0^2 (2x - x^2)\, dx.$

1 point: $A = \left[x^2 - \dfrac{x^3}{3}\right]_0^2 = 4 - \dfrac{8}{3} = \dfrac{4}{3}.$

(b) 2 points.

1 point: side $s = f(x) - g(x) = 2x - x^2$, so $V = \displaystyle\int_0^2 (2x - x^2)^2 dx.$

1 point: $\displaystyle\int_0^2 (4x^2 - 4x^3 + x^4)\, dx = \left[\tfrac{4}{3}x^3 - x^4 + \tfrac{x^5}{5}\right]_0^2 = \frac{32}{3} - 16 + \frac{32}{5} = \frac{16}{15}.$

(c) 2 points.

1 point: washers with $R = f(x) = 3x - x^2$ and $r = g(x) = x$ (the region lies between two curves above the axis).

1 point: $V = \pi \displaystyle\int_0^2 \left[(3x - x^2)^2 - x^2\right] dx.$

(d) 3 points.

1 point: The axis $y = 4$ is above the region. The farther boundary is g: $R = 4 - x$; the nearer is f: $r = 4 - (3x - x^2) = x^2 - 3x + 4.$

1 point: $V = \pi \displaystyle\int_0^2 \left[(4 - x)^2 - \left(x^2 - 3x + 4\right)^2\right] dx.$

1 point: $(4 - x)^2 - (x^2 - 3x + 4)^2 = -x^4 + 6x^3 - 16x^2 + 16x$, so $V = \pi \left[-\dfrac{x^5}{5} + \dfrac{3x^4}{2} - \dfrac{16x^3}{3} + 8x^2\right]_0^2 =$

$\pi \left(-\dfrac{32}{5} + 24 - \dfrac{128}{3} + 32\right) = \dfrac{104\pi}{15} \approx 21.782.$